

Received 30 April 2026, accepted 11 June 2026, date of publication 29 June 2026, date of current version 14 July 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3708195

RESEARCH ARTICLE

CoRF-Yoga: Cross-Modal Contrastive Learning for Robust, Feasible Yoga Pose Recognition

ZOLBOO DAMIRAN^{1,2}, (Graduate Student Member, IEEE),
TOMOKAZU MATSUI¹, (Member, IEEE), YUKI MATSUDA³, (Member, IEEE),
HIROHIKO SUWA¹, (Member, IEEE),
AND KEIICHI YASUMOTO¹, (Senior Member, IEEE)

¹Graduate School of Science and Technology, Nara Institute of Science and Technology, Nara 630-0192, Japan

²School of Information and Communication Technology, Mongolian University of Science and Technology, Ulaanbaatar 13341, Mongolia

³Faculty of Environmental, Life, Natural Science and Technology, Okayama University, Okayama 700-0082, Japan

Corresponding author: Zolboo Damiran (damiran.zolboo.cw5@naist.ac.jp)

This work was supported by JSPS KAKENHI under Grant JP25K03107.

ABSTRACT Recognizing yoga poses in real-world environments is challenging due to sensor-dependent noise, execution variability, and the high cost of large-scale manual annotation, which limits the scalability of existing supervised approaches. Prior works have predominantly relied on single-modality inputs and adopted either fully supervised or self-supervised contrastive learning paradigms. Fully supervised methods typically degrade when labeled data are scarce, while self-supervised contrastive approaches, although label-free, often struggle to learn sufficiently discriminative representations for fine-grained pose recognition. These limitations motivate supervised contrastive learning in a cross-modal setting to better exploit complementary information across modalities under practical sensing constraints. This paper presents *CoRF-Yoga*, a cross-modal contrastive learning framework for robust and label-efficient multimodal yoga pose recognition. CoRF-Yoga adopts a two-stage design consisting of contrastive pretraining to align RGB visual appearance and skeletal motion representations in a shared latent space, followed by frozen-encoder downstream classification with a lightweight MLP head. The framework supports both supervised contrastive learning (SCL) and self-supervised contrastive learning (Self-SCL) within the same architecture, enabling a controlled comparison of supervision levels. We evaluate CoRF-Yoga on *DSZ-Yoga*, a precisely synchronized RGB–motion dataset collected from twelve participants performing the *Sun Salutation B* sequence and consolidated into nine canonical yoga poses. Experiments across RGB-only, Motion-only, and RGB+Motion fusion settings demonstrate that contrastive pretraining substantially improves downstream recognition performance and cross-modal representation alignment. In particular, SCL achieves 99.61% accuracy for RGB+Motion fusion using only 25% labeled data under the LOSO protocol. In contrast, Self-SCL attains 66.10% fusion accuracy, indicating limitations of label-free contrastive pretraining for fine-grained pose recognition in this setting. These results demonstrate the importance of supervised contrastive learning for robust and label-efficient multimodal yoga pose recognition, while showing that cross-modal contrastive learning provides a practical foundation for efficient inference-time deployment in terms of latency and memory usage.

INDEX TERMS Cross-modal contrastive learning, human activity recognition, label efficiency, multimodal representation learning, supervised and self-supervised contrastive learning, wearable sensors, yoga pose recognition.

The associate editor coordinating the review of this manuscript and approving it for publication was Masood Ur-Rehman¹.

1. INTRODUCTION

Human Activity Recognition (HAR) supports a wide range of pervasive applications in health, sport, rehabilitation,

and fitness [1], [2]. The performance of HAR systems is significantly affected by the selected sensing modality. Vision-based systems achieve high accuracy by leveraging large volumes of pre-annotated training data and pre-trained models [3], [4]. However, these systems raise concerns regarding user privacy, increased deployment costs due to the requirement for cameras and sensors, and sensitivity to changes in illumination and occlusions. In contrast, wearable sensing with smartphone-based or inertial measurement unit (IMU)-based devices offers a more cost-effective and private solution. Nevertheless, such systems provide only limited motion information and typically require fully supervised learning approaches [5]. Recently, multimodal sensing that integrates visual and motion-based modalities has gained prominence in HAR research, as it enables the capture of complementary aspects of human movement [6], [7]. While multimodal systems can collect richer and more complementary information than single-modality approaches, they introduce additional practical challenges, including sensor synchronization, heterogeneous data fusion, and increased annotation effort across modalities [8].

To systematically evaluate recognition performance under these fine-grained conditions, we focus on the Sun Salutation B sequence [9]. This routine is widely practiced in Ashtanga yoga and consists of canonical poses that require precise body-level geometric alignment, making them sensitive to subtle variations in posture. Compared to simpler flows such as Sun Salutation A, Sun Salutation B introduces a richer set of standing and asymmetric postures (e.g., Chair and Warrior I), increasing pose-level ambiguity due to higher inter-class similarity among poses. At the same time, unlike more variable sequences such as Sun Salutation C, which often exhibit greater stylistic variation across instructors and traditions, Sun Salutation B remains highly consistent in pose ordering and execution. This consistency, illustrated in Figure 1, includes repeated poses within a single flow.

To address these challenges of fine-grained pose discrimination and limited supervision in multimodal yoga analysis, we introduce DSZ-Yoga, a precisely synchronized RGB–motion dataset capturing two complementary sensing modalities, including RGB video and full-body skeletal motion. We focus on RGB and wearable motion capture as a representative vision–wearable pair that is both practically deployable and scientifically complementary. RGB provides appearance cues, while motion provides explicit body kinematics. Aligning these modalities is valuable because RGB is sensitive to occlusion and viewpoint, whereas motion is more robust but lacks appearance and environmental context, allowing each modality to compensate for the other’s failure modes. Building on this dataset, we propose CoRF-Yoga, a cross-modal contrastive learning framework that learns shared, modality-invariant representations through cross-modal contrastive pretraining, followed by frozen-encoder downstream classification with a lightweight MLP head. We study two complementary contrastive learning paradigms: (i) *supervised contrastive learning (SCL)*, which



FIGURE 1. Illustration of the Sun Salutation B sequence considered in this study.

enforces pose-level semantic clustering using the SupCon loss, and (ii) *self-supervised contrastive learning (Self-SCL)*, which leverages temporal cross-modal correspondence and is optimized using the InfoNCE loss. Using a unified architecture isolates the effect of supervision level and clarifies the annotation-accuracy trade-off.

Through extensive experiments across different modality configurations and label budgets, we show that CoRF-Yoga enables effective cross-modal representation learning between RGB visual appearance and skeletal motion. In particular, we observe a clear cross-modal learning effect, where skeletal motion representations benefit substantially from alignment with RGB features during contrastive pretraining, leading to more stable and discriminative embeddings. While fully supervised models may achieve higher accuracy when abundant labeled data are available, CoRF-Yoga demonstrates competitive downstream performance under reduced labeling regimes, with especially strong gains for motion-based representations. Label-efficient cross-modal learning is essential for personalized home yoga coaching and scalable, instructor-independent platforms, where large labeled datasets are impractical. These results highlight the practical value of cross-modal contrastive learning for annotation-constrained yoga pose recognition. We adopt simple feature-level concatenation to keep fusion lightweight and to enable a controlled comparison, so that performance differences primarily reflect representation alignment rather than fusion-module capacity. In fused RGB+Motion recognition, SCL in CoRF-Yoga achieves 99.61 accuracy and attains the same performance using only 25% labeled data, improving over supervised training from scratch by 36.36 percentage points at the 25% label budget.

This work advances multimodal yoga pose recognition by improving cross-modal representation learning, robustness, and annotation efficiency under practical sensing constraints. Our main contributions are summarized as follows:

- 1) We propose *CoRF-Yoga*, a cross-modal contrastive learning framework that aligns RGB visual appearance and skeletal motion representations within a shared embedding space. The framework supports both supervised contrastive learning and self-supervised cross-modal contrastive learning within an identical architectural pipeline, enabling robust and transferable representations under heterogeneous sensing conditions.
- 2) We introduce *DSZ-Yoga*, a precisely synchronized RGB-motion dataset capturing the *Sun Salutation B* sequence. The dataset combines RGB video and full-body wearable motion capture with clip-level annotations, providing a controlled testbed for evaluating fine-grained cross-modal learning under realistic execution variability.
- 3) We empirically demonstrate a consistent cross-modal learning effect, where skeletal motion representations substantially improve under cross-modal contrastive pretraining and achieve more stable and discriminative embeddings than fully supervised unimodal baselines, particularly in low-label regimes.
- 4) We show that contrastive pretraining substantially improves annotation efficiency for yoga pose recognition. Supervised contrastive pretraining achieves near-saturated performance with significantly reduced labeled data, while self-supervised cross-modal contrastive pretraining provides consistent improvements over supervised baselines without requiring pose labels during representation learning.
- 5) We analyze trade-offs among representation quality, annotation cost, and computational efficiency. Results show that CoRF-Yoga improves robustness and accuracy with comparable model size and reduced end-to-end (E2E) inference latency and memory usage compared to end-to-end supervised baselines, offering practical guidance for scalable and privacy-preserving yoga coaching and HAR systems.

The remainder of this paper is organized as follows: Section II reviews related work in yoga pose recognition and multimodal representation learning. Section III describes the proposed dataset. Section IV presents the CoRF-Yoga framework. Section V reports experimental results across modalities and label regimes. Section VI discusses the implications of the results. Finally, Section VII concludes the paper.

II. RELATED WORK

A. YOGA POSE RECOGNITION AND DATASETS

Yoga pose recognition has been widely studied using vision-based and wearable sensing approaches [25].

However, most existing datasets and systems rely on single-modality inputs, particularly RGB video or images, and focus on recognizing isolated static postures rather than standardized pose sequences. Representative benchmarks such as Yoga-82 [10] enable supervised classification of predefined poses, but they often lack participant diversity and precisely synchronized multimodal recordings. While many studies adopt pose-estimation pipelines based on keypoint regression models such as OpenPose, MoveNet, or MediaPipe [11], [12], [13], [26], these approaches typically employ convolutional or recurrent neural networks to map visual or skeletal features to pose labels. As a result, models trained on such datasets often struggle to generalize to real-world scenarios involving sensor noise, viewpoint changes, or heterogeneous sensing conditions, especially when multiple sensing modalities are involved.

Consequently, existing yoga pose recognition studies remain largely constrained to single-modality inputs and fully supervised learning paradigms, with limited investigation into precisely synchronized multimodal sensing and label-efficient representation learning frameworks capable of capturing semantically rich, modality-invariant features for fine-grained yoga pose recognition.

B. CONTRASTIVE LEARNING FOR LABEL-EFFICIENT HAR

SCL (supervised contrastive learning) has emerged as an effective approach for improving label efficiency in HAR under limited supervision. By defining positive pairs based on shared class labels, supervised contrastive objectives explicitly impose class-level structure in the embedding space, leading to reduced intra-class variance and enhanced inter-class separability compared to conventional cross-entropy training [14]. Such class-aware representation learning is particularly beneficial for fine-grained recognition tasks. However, existing HAR studies predominantly investigate SCL in unimodal settings, such as RGB video or skeleton-based motion representations, and focus on coarse-grained activity categories, leaving its effectiveness in multimodal and cross-modal scenarios relatively underexplored.

Self-SCL further reduces annotation cost by learning transferable representations from unlabeled data. Foundational frameworks such as SimCLR [15] and MoCo [27], along with negative-free variants including BYOL [16], Barlow Twins [28], and VICReg [29], have demonstrated strong representation quality and label efficiency, motivating their adoption in HAR.

In HAR, contrastive learning has been extended to exploit temporal coherence, motion dynamics, and skeletal structure. Temporal contrastive objectives leverage temporally adjacent video clips [17], [18], while skeleton-based approaches use graph representations to learn view-invariant motion features [19], [20]. Although these methods consistently improve downstream performance under limited supervision, they remain predominantly unimodal and do not explicitly

TABLE 1. Summary of related work: strengths and limitations of existing approaches.

Research Area	Representative Examples	Strengths	Limitations
Yoga Pose Recognition & Datasets	Yoga-82 [10], yoga-specific methods [11]–[13]	Enables effective supervised classification of predefined poses.	Unimodal; limited diversity; no synchronized multimodal data; weak robustness to real-world conditions.
Supervised Contrastive Learning (SCL)	Khosla et al. [14]	Improves label efficiency; enhances intra-class compactness and inter-class separability.	Mostly unimodal; limited application to multimodal or fine-grained pose recognition.
Self-Supervised Contrastive Learning (Self-SCL)	SimCLR [15], BYOL [16], temporal/skeleton CL [17]–[20]	Learns representations without labels; captures temporal and motion dynamics.	Mostly unimodal; does not explicitly address cross-modal alignment; limited focus on fine-grained pose recognition.
Cross-Modal Contrastive Learning	RGB–IMU [21], IMU–video alignment [22]–[24]	Combines complementary modalities; robust to occlusion and sensor noise.	Focused on coarse-grained tasks; limited study of fine-grained pose recognition; lacks unified comparison of SCL vs. Self-SCL; limited analysis of annotation efficiency.

address cross-modal alignment for fine-grained pose recognition.

C. CROSS-MODAL CONTRASTIVE LEARNING FOR VISION AND WEARABLE SENSORS

Cross-modal representation learning aligns representations from heterogeneous sensing modalities to exploit complementary information and improve robustness across diverse sensing conditions. In HAR, prior work has increasingly focused on aligning vision with wearable sensing modalities. Studies combining RGB with IMU demonstrate that correlating visual context with motion dynamics improves robustness under visual occlusion or sensor noise [21]. More recent contrastive approaches treat synchronized vision–wearable pairs as natural positives, enabling representation learning without manual annotation. Related efforts further explore vision–motion alignment using IMU–video pairs or skeleton-based motion signals, showing improved transferability across sensing conditions [22], [23], [24].

Despite these advances, existing vision–wearable contrastive learning methods are predominantly evaluated on coarse-grained activity recognition tasks, where classes are defined by distinct temporal dynamics (e.g., walking or running). Fine-grained pose recognition poses substantially different challenges, as it requires precise spatial and semantic alignment to distinguish visually and kinematically similar postures. Moreover, many existing multimodal frameworks emphasize alignment or distillation under full supervision, with limited attention to annotation efficiency and systematic comparisons between supervised and self-supervised contrastive objectives within a unified pipeline.

Table 1 summarizes the strengths and limitations of existing approaches.

In contrast to prior work, this work focuses on fine-grained yoga pose recognition using precisely synchronized RGB–motion data and provides a unified evaluation of SCL and Self-SCL, explicitly analyzing the trade-off between representation quality and annotation cost.

III. DSZ-YOGA DATASET

A. OVERVIEW AND EXPERIMENTAL SETUP

DSZ-Yoga is a precisely synchronized multimodal dataset designed to support label-efficient cross-modal learning of fine-grained yoga poses under realistic execution conditions. The dataset captures complementary visual and kinematic information during natural yoga practice, enabling systematic investigation of multimodal representation learning under limited supervision.

Each participant completed two to three full repetitions of the *Sun Salutation B* sequence, introducing natural variation in execution speed, balance, and pose stability. The dataset comprises twelve volunteers (eight males and four females), aged 22–40 years (mean age 28.1), with heights ranging from 153 to 180 cm and weights from 48 to 75 kg, covering a broader range of physical characteristics. Only one participant had prior yoga experience, while the remaining participants were beginners without formal instruction; however, most participants reported regular engagement in physical activities such as gym training or sports. These heterogeneous characteristics introduce realistic variability in body proportions and execution styles, supporting evaluation under diverse user conditions. Detailed participant demographics are reported in Table 2.

During recording, the widely practiced *Sun Salutation B* sequence was performed as a continuous yoga flow. The nineteen ordered postures that constitute this sequence are summarized in Table 3. This sequence was selected because it provides a structured and continuous practice setting in which canonical poses are repeatedly executed under varying body orientations and execution styles, as shown in Figure 1. Although the sequence involves continuous motion, this study focuses on recognizing static yoga postures using short, fixed-length temporal clips extracted from the flow.

Repeated occurrences of identical postures within the sequence were preserved during annotation to retain natural execution variability. For downstream classification experiments, these instances were consolidated into nine target pose

TABLE 2. Participant demographics in the DSZ-Yoga dataset.

ID	Gender	Age	Height (cm)	Weight (kg)	Yoga Experience
Y001	Female	25	159	48	None
Y002	Male	22	180	65	None
Y003	Male	26	173	63	None
Y004	Male	25	172	55	None
Y005	Male	37	179	70	>1 yr (2–3/week)
Y006	Male	23	176	63	None
Y007	Female	24	165	51	None
Y008	Male	37	163	62	None
Y009	Male	40	165	70	None
Y010	Female	28	153	54	None
Y011	Female	22	164	53	None
Y012	Male	28	172	75	None

TABLE 3. Canonical Sun Salutation B sequence with pose labels.

#	Sanskrit	English
1	Tadasana	Standing Mountain (Start)
2	Utkatasana	Chair
3	Uttanasana	Standing Forward Bend
4	Ardha Uttanasana	Half Forward Bend
5	Chaturanga Dandasana	Half-Plank
6	Urdhva Mukha Svanasana	Upward Facing Dog
7	Adho Mukha Svanasana	Downward Facing Dog
8	Virabhadrasana I (R)	Warrior I (Right)
9	Chaturanga Dandasana	Half-Plank
10	Urdhva Mukha Svanasana	Upward Facing Dog
11	Adho Mukha Svanasana	Downward Facing Dog
12	Virabhadrasana I (L)	Warrior I (Left)
13	Chaturanga Dandasana	Half-Plank
14	Urdhva Mukha Svanasana	Upward Facing Dog
15	Adho Mukha Svanasana	Downward Facing Dog (Hold)
16	Ardha Uttanasana	Half Forward Bend
17	Uttanasana	Standing Forward Bend
18	Utkatasana	Chair
19	Tadasana	Standing Mountain (End)

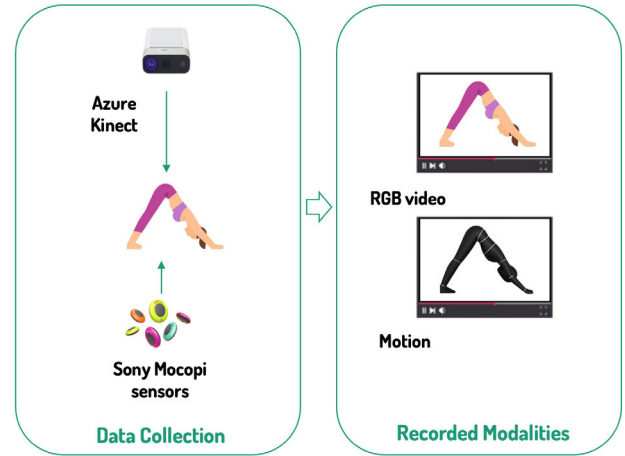
classes: Standing Mountain, Chair, Standing Forward Bend, Half Forward Bend, Half-Plank, Upward Facing Dog, Downward Facing Dog, Warrior I (Right), and Warrior I (Left). This formulation enables consistent clip-level supervision while maintaining substantial intra-class variability across multiple executions of each pose.

Detailed descriptions of the sensing configuration, temporal alignment, annotation protocol, and dataset statistics are provided in the following subsections.

B. SENSOR CONFIGURATION AND TEMPORAL ALIGNMENT

The DSZ-Yoga dataset integrates two complementary sensing modalities, RGB vision and wearable motion capture, to capture visual appearance and body kinematics with short-term motion context during yoga performance. The sensing setup is illustrated in Figure 2. A single Azure Kinect camera was positioned at a frontal viewpoint to record RGB video. Full-body kinematic information was captured using twelve Sony Mocopi sensors (Pro Kit), whose raw orientation data

were converted into 3D joint coordinates through forward kinematics. The resulting motion representation consists of a fixed set of 3D joint positions in a consistent skeletal topology.

**FIGURE 2.** Multimodal data collection setup with a single Azure Kinect camera and twelve Sony Mocopi sensors for full-body motion capture.

All sensors operated concurrently and recorded device-specific timestamps, which were later standardized during preprocessing. Kinect metadata provided per-frame host-wall timestamps for the RGB stream. For motion capture, timestamps were reconstructed by deriving the recording start time from the end-of-recording timestamp embedded in the .bvh filename and the sequence duration. To verify the reliability of the reconstructed motion timestamps, we visually inspected the resulting 3D skeletal motion sequences. The joint trajectories exhibited smooth temporal evolution and coherent pose transitions across the sequence, indicating that the derived timestamps preserve the temporal structure of the original motion recordings and are sufficient for clip-level synchronization.

No spatial extrinsic calibration was performed, as the proposed framework aligns modalities temporally and learns cross-modal representations at the feature level without requiring a shared geometric coordinate system. This design choice avoids dependence on precise sensor placement and supports flexible integration of heterogeneous sensing modalities.

Although the sensing devices originally produced heterogeneous raw formats (e.g., RGB as .png, and motion capture as .bvh), all streams were converted into standardized HDF5 files containing signal references and synchronized timing information. This standardization supports consistent timestamp alignment and efficient multimodal loading across all experiments. Table 4 summarizes the sensing devices and standardized file formats used in DSZ-Yoga.

C. DATA ACQUISITION AND SYNCHRONIZATION

All sensing devices operated concurrently during recording, but each modality produced timestamps according to its own device clock. To enable multimodal learning, all streams were

TABLE 4. Sensing modalities and standardized file formats in the DSZ-Yoga dataset.

Modality	Device	Standardized Files
RGB (single view)	Azure Kinect	<code>rgb.h5</code> (image paths + host-wall timestamps)
Motion (12 sensors)	Sony Mocopi	<code>motion.h5</code> (3D joint positions + timestamps), <code>joint_names</code>

aligned to a unified global timeline during preprocessing. The synchronization pipeline consists of three stages.

1) STANDARDIZATION OF RAW SENSOR STREAMS

Raw data from each device were converted into structured HDF5 files that store both signal content and timing information. The RGB stream uses host-wall timestamps extracted from the Kinect `metadata.h5` files, while Mocopi motion timestamps were reconstructed by deriving the recording start time from the end-of-recording timestamp embedded in the `.bvh` filename and the sequence duration. This standardization produces consistent microsecond-level timestamps for all modalities and enables efficient parallel access for later processing. Although Azure Kinect also provides depth data, depth streams were not used in this study because RGB and motion capture offer fundamentally different and complementary characteristics. RGB captures visual appearance, whereas motion capture provides explicit body kinematics, while depth largely overlaps with RGB in geometric information and offers limited additional benefit for the targeted cross-modal learning setting.

2) SESSION-LEVEL ALIGNMENT AND CLIP INDEXING

For each recording session, the intersection of valid timestamp ranges across RGB and motion streams was computed to identify the interval in which all modalities were simultaneously available. Within this aligned interval, fixed-length clips were defined using a 2.0 s window and a 1.0 s hop. Each clip was assigned a pose label based on its absolute start time and the ground-truth annotation intervals of the Sun Salutation B sequence. This stage produces a global index of synchronized multimodal clips for downstream training and evaluation.

3) FRAME-LEVEL RESAMPLING AND MULTIMODAL CLIP CONSTRUCTION

To construct training-ready multimodal samples, a reference clock was generated for each clip at 8 Hz, yielding 16 uniformly spaced timestamps over the 2.0 s window. For each timestamp, the nearest RGB and motion frames were retrieved from their respective HDF5 files. RGB images were resized to 224×224 pixels, and motion data were temporally interpolated to the reference timeline. The resulting clip, containing frame-level synchronized representations from both modalities, was stored as a compressed `.npz` file for efficient loading.

This synchronization pipeline enables reliable clip-level temporal alignment across heterogeneous sensors without requiring spatial calibration, forming the basis for robust cross-modal representation learning in the proposed CoRF-Yoga framework.

D. ANNOTATION PROTOCOL

The Sun Salutation B sequence consists of nineteen ordered postures, each with well-defined temporal boundaries. To obtain ground-truth pose intervals, all RGB recordings were manually annotated using the VGG Image Annotator (VIA) [30] (version 3.0.13), a lightweight web-based tool that supports precise temporal labeling of video streams. Annotators marked the start and end times of each posture by visually inspecting the frontal RGB view, ensuring consistency across sessions. The resulting annotations were exported in VIA JSON format for subsequent preprocessing.

The VIA annotations provide pose intervals in relative video time. During preprocessing, these intervals were converted into absolute timestamps by aligning them with the absolute start time of the RGB stream, derived from the Kinect host-wall timestamps. This conversion enabled each pose interval to be matched directly to the absolute clip timestamps computed during multimodal synchronization. Although the Sun Salutation B sequence contains repeated occurrences of several canonical postures (e.g., Half-Plank or Downward Facing Dog), these repetitions were preserved during annotation but later consolidated into nine distinct pose classes for downstream classification experiments.

The resulting annotation files provide frame-accurate temporal segmentation for each participant and constitute the supervisory signal used for clip-level labeling and downstream evaluation.

To strengthen dataset credibility, we conducted an annotation verification procedure. Specifically, one subject (Y012) was independently re-annotated, and the resulting annotations were compared with the original labels at the segment level. A segment was considered consistent if both the pose label and temporal boundaries matched within a tolerance of ± 1.0 second. All 19 segments were found to be consistent, yielding an agreement rate of 100%. While this verification was limited to a subset of the dataset, the result indicates high annotation consistency and supports the overall reliability of the dataset.

E. DATASET COMPOSITION AND STATISTICS

After preprocessing, synchronization, and quality filtering, DSZ-Yoga contains 2945 multimodal clips, of which 2071 provide valid, temporally aligned RGB and motion data and overlap at least one annotated pose interval. The remaining clips were excluded because they overlapped only transition segments and therefore could not be assigned a pose label with sufficient confidence. Each clip spans 2.0 s and corresponds to a temporally synchronized RGB-motion segment extracted from continuous yoga practice.

TABLE 5. Clip counts per pose class in the DSZ-Yoga dataset.

Pose Class	# Clips
Standing Mountain	235
Chair	175
Standing Forward Bend	209
Half Forward Bend	240
Half-Plank	264
Upward Facing Dog	353
Downward Facing Dog	347
Warrior I (Right)	128
Warrior I (Left)	120
Total	2071

Although the Sun Salutation B sequence contains nineteen canonical posture steps, repeated occurrences of identical postures were merged into nine distinct pose classes for downstream classification experiments. The distribution of the 2071 synchronized clips across these classes is summarized in Table 5.

While the participant pool is limited in size and gender balance, the dataset¹ emphasizes multimodal diversity and execution variability and is intended primarily for methodological evaluation rather than population-level generalization. Class labels, clip boundaries, and modality-aligned timestamps are stored in a global index file used by all training and evaluation procedures. Distinct from large-scale web-scraped datasets, DSZ-Yoga focuses on high-fidelity multimodal synchronization and realistic execution variability. This design choice ensures dense, temporally aligned supervision across two modalities, making it suitable for studying label-efficient and cross-modal representation learning within the proposed CoRF-Yoga framework.

In summary, DSZ-Yoga provides temporally synchronized RGB-motion clips with substantial intra-class variability arising from repeated executions within a continuous yoga flow. These characteristics make the dataset a suitable testbed for studying cross-modal representation learning and label-efficient pose recognition under limited annotations. Motivated by this setting, we next introduce *CoRF-Yoga*, a unified framework for supervised and self-supervised cross-modal contrastive pretraining and frozen-encoder downstream evaluation.

IV. PROPOSED FRAMEWORK: CoRF-YOGA

This section presents CoRF-Yoga, a cross-modal contrastive learning framework for multimodal yoga pose recognition using synchronized RGB video and skeletal motion data. The overall framework, learning objectives, and downstream usage of the learned representations are described in the following subsections.

A. FRAMEWORK OVERVIEW

Figure 3 illustrates the overall architecture of the proposed *CoRF-Yoga* framework under the RGB-motion setting. The

¹The DSZ-Yoga dataset is available at <https://zenodo.org/records/18297451>.

framework is designed to learn robust and label-efficient representations for yoga pose recognition by aligning heterogeneous sensing modalities through contrastive learning.

CoRF-Yoga follows a two-stage training paradigm. In the first stage, modality-specific encoders are trained using contrastive objectives to learn a shared representation space across RGB visual appearance and skeletal motion. In the second stage, the pretrained encoders are frozen and reused as fixed feature extractors for downstream supervised pose classification.

During contrastive pretraining (Figure 3A), temporally synchronized RGB frames and skeletal motion sequences are processed independently by dedicated encoders. The RGB branch employs a ResNet-based image encoder operating on single RGB frames extracted from synchronized clips. Each encoder produces a compact modality-specific representation, which is further mapped into a shared latent space via lightweight projection heads. Depending on the training setting, these representations are optimized using either supervised contrastive learning, where pose labels define positive pairs, or self-supervised cross-modal contrastive learning, where positive pairs are defined by temporal alignment across modalities. Aside from the contrastive loss formulation, the architectural pipeline remains identical across both settings.

During downstream pose classification (Figure 3B), the projection heads are discarded and the pretrained encoders remain frozen. A lightweight multilayer perceptron (MLP) classification head is trained on labeled data, while encoder parameters are kept fixed. For each input clip, RGB and motion features are extracted independently and fused at the feature level using simple concatenation. The fused representation is then passed to the MLP classifier to predict the yoga pose label. This design decouples representation learning from classification, enabling effective learning under limited annotation budgets.

Because each modality is encoded independently and fusion occurs at the feature level, CoRF-Yoga naturally supports inference using any available subset of modalities. In this paper, we evaluate RGB-only, motion-only, and RGB+motion settings; extension to additional modalities follows the same design. This flexibility allows the framework to operate under partial sensor availability without requiring architectural changes. Overall, the two-stage design of CoRF-Yoga provides a unified and extensible framework for evaluating supervised and self-supervised cross-modal contrastive learning for robust and label-efficient yoga pose recognition.

B. MODALITY-SPECIFIC DATA AUGMENTATION

During contrastive pretraining, CoRF-Yoga applies modality-specific data augmentations to generate two augmented views per modality from each synchronized clip. For RGB inputs, standard spatial and photometric perturbations are applied to frames from a single fixed camera view, while skeletal motion is augmented using geometric and temporal perturbations.

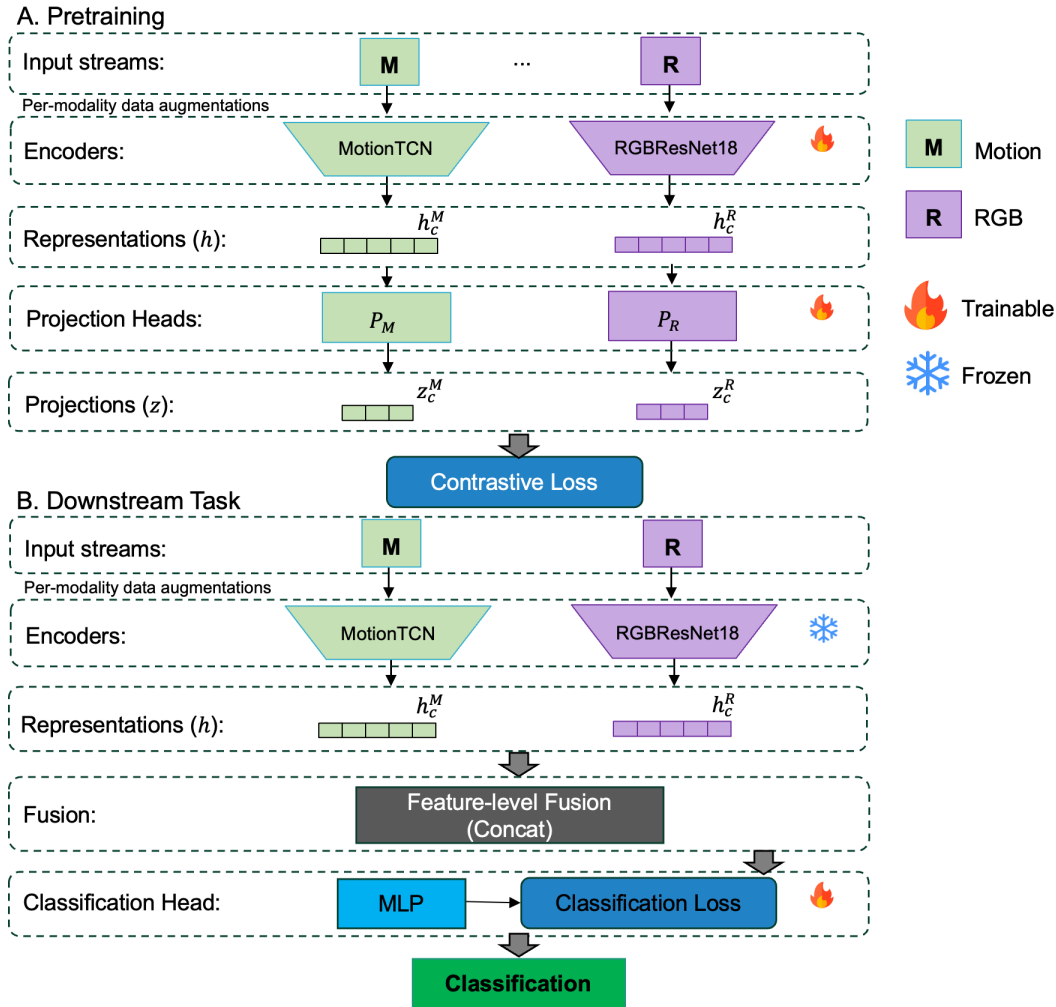


FIGURE 3. Overall architecture of the CoRF-Yoga framework. (A) During cross-modal contrastive pretraining, temporally synchronized RGB and skeletal motion inputs are processed by modality-specific encoders and aligned in a contrastive shared latent space using contrastive learning. (B) During downstream pose classification, the pretrained encoders are frozen and paired with a lightweight MLP classifier for label-efficient yoga pose recognition.

Cross-modal temporal alignment is preserved throughout augmentation.

C. MODALITY-SPECIFIC ENCODERS

CoRF-Yoga employs lightweight, modality-specific encoders tailored to RGB visual appearance and skeletal motion. Each modality is encoded independently to produce a compact latent representation, enabling modality-specific feature extraction while supporting cross-modal alignment through contrastive learning.

1) RGB ENCODER

The RGB modality is encoded using a ResNet18 backbone, selected for its favorable balance between representational capacity and computational efficiency. To reduce computational overhead and avoid redundant temporal modeling, the encoder operates on a single representative frame extracted from each synchronized clip, while temporal dynamics are captured by the skeletal motion modality. The encoder

outputs a 512-dimensional appearance feature vector. This design choice is empirically validated under the LOSO protocol, where motion-only models consistently outperform RGB-only models. This indicates that temporal information is effectively captured by the motion modality, while the RGB branch provides complementary spatial cues.

2) MOTION (SKELETON) ENCODER

Skeletal motion is encoded using a temporal convolutional network (TCN) operating on joint positions and velocities. The TCN is chosen for its ability to capture short-term temporal dynamics in structured motion data with low computational cost, producing a 512-dimensional motion representation.

D. PROJECTION HEADS AND SHARED LATENT SPACE

To facilitate contrastive alignment between RGB visual appearance and skeletal motion modalities, CoRF-Yoga maps the output of each modality-specific encoder into a shared

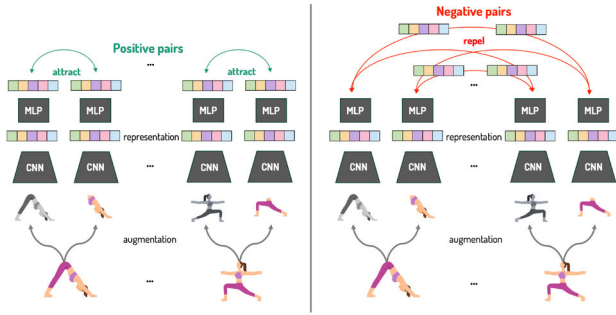


FIGURE 4. Conceptual illustration of contrastive learning. Positive pairs are pulled together in the latent space, while negative pairs are pushed apart, under both supervised (label-based) and self-supervised cross-modal (synchronization-based) settings.

latent representation space using lightweight projection heads. These projection heads follow a SimCLR-style multilayer perceptron (MLP) design and are used exclusively during the contrastive pretraining stage for both supervised and self-supervised contrastive objectives.

Given a synchronized input clip c , the encoder for modality $s \in \{\text{rgb}, \text{motion}\}$ produces a representation $\mathbf{h}_c^s \in \mathbb{R}^{512}$ (corresponding to the “Representations (h)” stage in Figure 3). Each encoder output is then mapped to a lower-dimensional projection $\mathbf{z}_c^s \in \mathbb{R}^{128}$ (“Projections (z)” in Figure 3) via a lightweight multilayer perceptron (MLP) with batch normalization and nonlinear activations. The projected features are subsequently ℓ_2 -normalized to produce unit-length embeddings:

$$\mathbf{z}_c^s = \frac{g_s(\mathbf{h}_c^s)}{\|g_s(\mathbf{h}_c^s)\|_2}.$$

Following common practice in contrastive learning frameworks such as SimCLR, the projection heads are used exclusively during contrastive pretraining and are discarded afterward. Although the contrastive objective is optimized in the projection space $\mathbf{z}$, the encoder representations $\mathbf{h}$ are retained for all downstream tasks, including frozen-encoder pose classification and feature-level fusion. Accordingly, downstream evaluation and visualization (e.g., UMAP) are performed in the encoder space $\mathbf{h}$, while cross-modal retrieval is evaluated in the projection space $\mathbf{z}$, where the contrastive alignment is explicitly enforced.

By mapping modality-specific encoder outputs into a common 128-dimensional latent space, the projection heads enable direct comparison of heterogeneous embeddings using cosine similarity. This shared latent space forms the basis for both supervised contrastive learning and self-supervised cross-modal contrastive learning objectives, allowing synchronized RGB and motion representations to be aligned effectively.

E. CONTRASTIVE LEARNING OBJECTIVES

CoRF-Yoga supports two contrastive pretraining settings within the same architectural pipeline, namely SCL and Self-SCL. Both settings operate on modality-specific

projections in a shared latent space and use identical encoder architectures, data augmentation strategies, and optimization settings. The key difference lies in how positive pairs are defined; in SCL, positives are determined using pose labels across modalities and augmented views, whereas in Self-SCL, positives are defined solely by temporal synchronization between RGB and motion modalities without using annotations. Aside from the contrastive loss formulation, the training pipeline remains unchanged across the two settings. Figure 4 provides a conceptual illustration of contrastive learning and is not intended to depict a specific network architecture.

1) SUPERVISED CONTRASTIVE LEARNING

In the supervised contrastive variant, pose annotations are used to explicitly define positive pairs during pretraining. Given a batch of B labeled RGB–motion clips, each clip is augmented to produce two views per modality. We construct a unified embedding set by collecting all projected representations across modalities and augmented views within the batch. Within this set, all embeddings that share the same pose label are treated as positives, while embeddings from different pose classes serve as negatives.

Let $\mathbf{z}_i \in \mathbb{R}^{128}$ denote the ℓ_2 -normalized projection of the i -th embedding in the unified set (each embedding originates from either the RGB or motion modality and from one of the two augmented views). Following the SCL formulation, the loss for an anchor embedding $\mathbf{z}_i$ is defined as

$$\mathcal{L}_i^{\text{SupCon}} = -\frac{1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_p)/\tau)}{\sum_{a \in \mathcal{A}(i)} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_a)/\tau)}, \quad (1)$$

where $\mathcal{P}(i)$ denotes the set of indices of all positive embeddings sharing the same pose label as anchor i , $\mathcal{A}(i)$ denotes the set of all other embeddings in the batch excluding the anchor itself, $\text{sim}(\cdot, \cdot)$ is cosine similarity, and τ is a temperature parameter.

2) SELF-SUPERVISED CROSS-MODAL CONTRASTIVE LEARNING

Positive pairs are defined based on temporal synchronization across modalities, where RGB and skeletal motion embeddings extracted from the same clip are treated as positives, while embeddings from different clips within the batch act as negatives.

Let $\mathbf{z}_{c,k}^{\text{rgb}}$ and $\mathbf{z}_{c,k}^{\text{motion}}$ denote the ℓ_2 -normalized projected embeddings obtained from the k -th augmented view ($k \in \{1, 2\}$) of the RGB and motion modalities for clip c . For a batch of B synchronized clips, the cross-modal InfoNCE loss for a given augmented view k is defined as

$$\mathcal{L}_{\text{cross}}^{(k)} = -\frac{1}{B} \sum_{c=1}^B \log \frac{\exp(\text{sim}(\mathbf{z}_{c,k}^{\text{rgb}}, \mathbf{z}_{c,k}^{\text{motion}})/\tau)}{\sum_{c'=1}^B \exp(\text{sim}(\mathbf{z}_{c,k}^{\text{rgb}}, \mathbf{z}_{c',k}^{\text{motion}})/\tau)}, \quad (2)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and τ is a temperature parameter. Here, each RGB embedding $\mathbf{z}_{c,k}^{\text{rgb}}$ acts as

an anchor and is contrasted against all motion embeddings $\{\mathbf{z}_{c',k}^{\text{motion}}\}_{c'=1}^B$ in the batch.

In practice, the cross-modal loss is computed symmetrically in both RGB→motion and motion→RGB directions and averaged across the two augmented view pairs ($k = 1, 2$). In addition to the cross-modal objective, CoRF-Yoga applies intra-modal contrastive regularization by computing InfoNCE losses between the two augmented views of the same modality (RGB–RGB and motion–motion). The final self-supervised objective is a weighted combination of the cross-modal and intra-modal losses, where the relative contributions are controlled by scalar weights. In the final configuration used in this study, equal weights are assigned to all terms, and the intra-modal weight is fixed at $\lambda = 1.0$ throughout training. Experimental results from the ablation study (see Appendix) demonstrate that a fixed intra-modal weight of $\lambda = 1.0$ yields the best overall performance across label fractions; accordingly, it is adopted as the default configuration.

By jointly encouraging cross-modal alignment and intra-modal consistency, the self-supervised objective learns modality-invariant yet modality-consistent representations without requiring pose annotations. These learned embeddings form the basis for the downstream pose classification tasks described in Section IV-G.

F. CROSS-MODAL CONTRASTIVE PRETRAINING PROCEDURE

During cross-modal contrastive pretraining, batches of temporally synchronized RGB and skeletal motion clips are augmented to generate two views per modality. Each view is processed by its corresponding modality-specific encoder and projected into a shared latent space, where the representations are optimized using either the supervised or self-supervised contrastive objective described in Section IV-E. Encoder and projection-head parameters are trained jointly, and the resulting pretrained encoders are subsequently evaluated using a frozen-encoder downstream classification protocol. Further implementation details follow the experimental setup described in Section V.

G. DOWNSTREAM POSE CLASSIFICATION

After contrastive pretraining, CoRF-Yoga evaluates the learned representations through a supervised downstream yoga pose classification task. In all experiments, the pretrained modality-specific encoders are frozen and reused as fixed feature extractors, while only a lightweight classification head is trained on labeled data. This protocol isolates the effect of representation quality learned during pretraining and enables a fair comparison between supervised and self-supervised contrastive methods.

1) FROZEN FEATURE EXTRACTION

Given an input clip, each available modality is processed independently by its corresponding pretrained encoder,

whose parameters remain fixed throughout downstream training.

2) CLASSIFIER TRAINING

The resulting feature representation is passed to a lightweight MLP classifier consisting of two fully connected layers with batch normalization, ReLU activation, and dropout. Only the classifier parameters are optimized using cross-entropy loss, while the encoder weights remain fixed. Optimization is performed using AdamW with a cosine learning-rate schedule.

3) EVALUATION PROTOCOL

Downstream performance is evaluated using a leave-one-subject-out (LOSO) cross-validation protocol. To assess label efficiency, only the downstream classifier training stage is repeated under varying label fractions (10%, 25%, 50%, 75%, and 100% of the training data), while the pretrained encoders remain unchanged and the test subject remains fixed for each LOSO fold. For each fold and label fraction, the model is trained on the reduced training set and evaluated on a held-out subject. Final results are reported as the mean and standard deviation across LOSO folds.

This frozen-encoder downstream protocol ensures a controlled and reproducible evaluation of the representations learned through supervised and self-supervised cross-modal contrastive pretraining.

H. FEATURE-LEVEL FUSION AND PARTIAL-MODALITY INFERENCE

CoRF-Yoga integrates information from multiple sensing modalities using a lightweight feature-level fusion strategy. Rather than performing early fusion at the input level or late fusion at the decision level, the framework combines modality-specific representations after independent encoding, allowing each encoder to specialize in extracting modality-relevant features while enabling effective multimodal integration.

1) FUSION STRATEGIES

a: FEATURE-LEVEL FUSION

Given pretrained and frozen encoders, CoRF-Yoga performs fusion by concatenating modality-specific feature vectors at the representation level. For the RGB–motion setting, the fused feature vector is constructed as

$$\mathbf{h}_{\text{fused}} = [\mathbf{h}_{\text{rgb}}; \mathbf{h}_{\text{motion}}],$$

where $\mathbf{h}_{\text{rgb}}$ and $\mathbf{h}_{\text{motion}}$ denote the encoder outputs for the RGB and skeletal motion modalities, respectively. This simple concatenation is adopted to preserve the pretrained modality-specific representations without distortion. Introducing additional fusion parameters or transformations could alter the geometry of the contrastively learned embedding space and obscure the effects of pretraining. By directly concatenating the frozen RGB and motion features, the

downstream classifier operates on the original representations learned during contrastive pretraining, enabling a clearer assessment of representation quality while maintaining compatibility between modalities.

b: GATED FUSION

In addition to simple concatenation, we also consider a gated fusion strategy that adaptively combines modality features:

$$\mathbf{g} = \sigma(\mathbf{W}_g[\mathbf{h}_{\text{rgb}}; \mathbf{h}_{\text{motion}}])$$

$$\mathbf{h}_{\text{fused}} = \mathbf{g} \odot \mathbf{h}_{\text{rgb}} + (1 - \mathbf{g}) \odot \mathbf{h}_{\text{motion}}$$

where $\sigma(\cdot)$ denotes the sigmoid function, $\odot$ represents element-wise multiplication, and $\mathbf{g}$ has the same dimensionality as the modality features. This mechanism allows the model to adaptively weight RGB and motion features on a per-sample basis, rather than relying on a fixed combination.

2) PARTIAL-MODALITY INFERENCE

Because each modality is encoded independently and fusion is performed at the feature level, CoRF-Yoga supports inference under varying modality availability through model selection. When only a unimodal input is available (e.g., RGB-only or motion-only), the system falls back to the corresponding unimodal classifier trained on that modality. When both RGB and motion modalities are available, their feature representations are concatenated and classified using the fused RGB+motion model. This design enables flexible deployment under partial sensor availability without requiring architectural changes or retraining.

Overall, the feature-level fusion strategy in CoRF-Yoga provides a simple yet effective mechanism for multimodal integration while preserving robustness to partial sensor availability.

V. EXPERIMENTS AND RESULTS

This section evaluates the proposed CoRF-Yoga framework under different supervision levels. In addition to SCL, we include Self-SCL to pretrain encoders under fully unlabeled conditions where pose annotations are unavailable. This enables a direct comparison between supervised and label-free contrastive pretraining within the same pipeline, clarifying the trade-off between annotation cost and downstream performance.

A. EXPERIMENTAL SETUP

1) DATASET AND EVALUATION PROTOCOL

All experiments use the DSZ-Yoga clips described in Section III (2.0 s windows, 1.0 s hop, 8 Hz sampling, and 9 pose classes). We evaluate three modality configurations: (1) RGB-only, (2) Motion-only, and (3) feature-level RGB+Motion fusion. For fusion, modality-specific embeddings extracted by the encoders are concatenated prior to classification (Section IV-H).

We adopt a leave-one-subject-out (LOSO) cross-validation protocol to evaluate cross-subject generalization. In each

fold, all clips from one subject are held out as the test set, while the remaining subjects are used for training. This process is repeated across all subjects, and the final performance is reported as the average over all folds. This protocol ensures that there is no subject overlap between training, validation, and test sets, providing a realistic evaluation of generalization to unseen individuals.

To study label efficiency, the training set in each LOSO fold is further subsampled using label fractions of 10%, 25%, 50%, 75%, and 100% with class-balanced sampling. From the remaining subjects, one subject is randomly selected as an independent validation subject for model selection, while the others are used for training. All LOSO folds, validation-subject selection, and label-fraction subsampling are generated with random shuffling using a fixed seed for reproducibility.

2) COMPARED METHODS

We compare CoRF-Yoga against end-to-end supervised baselines under the same modality configurations described above. For the RGB baseline, a ResNet-18 backbone initialized with ImageNet pretrained weights is trained jointly with an MLP classification head using cross-entropy loss. For the Motion baseline, a temporal convolutional network (TCN) is trained from random initialization using pose labels. For the Fusion baseline, RGB and Motion features extracted by their respective encoders are concatenated and classified by an MLP head, with all parameters trained end-to-end. This asymmetric initialization may introduce modality bias, potentially favoring RGB representations and affecting the fairness of baseline comparisons.

CoRF-Yoga uses the same encoder architectures but first performs contrastive pretraining using either SCL, optimized with the SupCon loss, or Self-SCL, optimized with the InfoNCE loss. During downstream evaluation, the pretrained encoders are frozen and only a lightweight MLP classifier head is optimized (Section IV-G). All methods follow the same cross-validation folds, label-fraction subsampling, preprocessing pipeline, and evaluation metrics to ensure fair comparison.

Learning rates for supervised baselines are selected per modality, reflecting differences in encoder initialization (ImageNet-pretrained RGB versus randomly initialized motion), while a unified learning rate is used for frozen downstream evaluation.

We emphasize that the supervised baseline represents conventional end-to-end supervised learning. In contrast, CoRF-Yoga performs contrastive pretraining using either supervised or self-supervised contrastive learning to learn representations, and is then evaluated using a frozen-encoder downstream protocol where only an MLP head is trained. Cross-modal retrieval is computed directly after pretraining using the learned embedding space.

We report results for supervised contrastive pretraining first, followed by self-supervised pretraining to highlight the trade-off between label cost and performance.

TABLE 6. Comparison with representative state-of-the-art methods. Results are not directly comparable due to differences in datasets, modalities, and evaluation protocols. DSZ-Yoga is the first benchmark pairing RGB and motion modalities for yoga pose recognition.

Method	Year	Modality	Learning	Dataset	Protocol	Acc (%)
Yoga-82 [10]	2020	RGB	Supervised	Yoga-82	Train/Test split	74.91
ViT [31]	2021	RGB (Image Patches)	Supervised	ImageNet	ImageNet split	88.55 [†]
ST-GCN [32]	2018	Skeleton (Graph-based)	Supervised	NTU-RGB+D	X-Sub / X-View	81.5 / 88.3
2s-AGCN [33]	2019	Skeleton (Graph-based)	Supervised	NTU-RGB+D	X-Sub / X-View	88.5 / 95.1
PoseC3D [34]	2022	Skeleton (3D heatmap)	Supervised	NTU-RGB+D 60	X-Sub / X-View	94.1 / 97.1
MAF-Net [35]	2025	RGB + Skeleton	Supervised (Fusion)	NTU-RGB+D	X-Sub / X-View	90.6 / 97.3
CoRF-Yoga (Ours)	2026	RGB + Motion	SCL	DSZ-Yoga	LOSO	99.66

No existing method has been evaluated on DSZ-Yoga; the listed methods provide contextual comparison across related domains. Our method additionally supports cross-modal retrieval.

Protocol definitions: Train/Test split: predefined fixed split provided by the dataset; ImageNet split: standard training/validation split used in ImageNet; X-Sub: cross-subject evaluation (train/test subjects are disjoint); X-View: cross-view evaluation (train/test camera views are disjoint); LOSO: leave-one-subject-out cross-validation.

[†] Pre-trained on JFT-300M

TABLE 7. Training hyperparameters for supervised baselines and frozen-encoder downstream evaluation.

Setting	Baseline (End-to-End)	Downstream (Frozen Encoder)
Optimizer	AdamW	AdamW
Learning rate	RGB, Fusion: 1×10^{-4} Motion: 1×10^{-3}	1×10^{-3}
Weight decay	0.01	0.01
Scheduler	Cosine Annealing	Cosine Annealing
Epochs	30	30
Batch size	32	32
Trainable parameters	Encoder + MLP	MLP only
Initialization	RGB: ImageNet Motion: Random	Contrastively pretrained (frozen)

To further contextualize the proposed approach, we compare it with representative recent state-of-the-art methods in human action and yoga pose recognition, including skeleton-based, RGB-based, and multimodal approaches. These methods are widely adopted in the literature and provide a broader reference for evaluating the effectiveness of cross-modal contrastive learning. A summary of the comparison is presented in Table 6.

As shown in Table 6, existing methods are typically limited to single-modality settings or evaluated under different datasets and protocols, whereas the proposed CoRF-Yoga framework addresses multimodal learning with cross-modal alignment on the DSZ-Yoga benchmark.

Downstream pose recognition is evaluated using the frozen-encoder protocol described in Section IV. Key training hyperparameters and the experimental environment are summarized in Tables 7 and 8, respectively.

3) EVALUATION METRICS

We evaluate downstream pose recognition, cross-modal alignment, and computational efficiency.

a: DOWNSTREAM CLASSIFICATION

Performance is reported using accuracy and macro-F1 on the held-out test split of each fold. Macro-F1 is used to account for class imbalance across the nine pose classes.

TABLE 8. Experimental environment.

Component	Details
Processor	Intel Core i7-13700
GPU	NVIDIA GeForce RTX 4080
RAM	64 GB
Software	Python 3.11.13
	PyTorch 2.5.1
	CUDA 12.1

b: REPRESENTATION LEARNING (CROSS-MODAL RETRIEVAL).

To assess cross-modal alignment, we evaluate pose-level cross-modal retrieval using Recall@1 (R@1), Recall@5 (R@5), Recall@10 (R@10), and mean average precision (mAP).

A query embedding from one modality is considered correct if at least one of the top- K retrieved samples in the other modality (ranked by cosine similarity) shares the same pose label; retrieval is evaluated in the projection space z unless stated otherwise.

c: COMPUTATIONAL EFFICIENCY

We measure inference latency, parameter count (M), and peak GPU memory usage (GB). Latency is reported as the mean forward-pass time over 100 iterations using synthetic inputs of fixed dimensions, excluding I/O overhead and including a warm-up phase to account for GPU initialization. All efficiency measurements are conducted using the same experimental environment described in Table 8.

B. SUPERVISED BASELINES

We evaluate fully supervised end-to-end baselines under the same modality configurations. Table 9 reports the downstream pose recognition performance. Across modalities, feature-level fusion consistently outperforms unimodal RGB and Motion models, indicating that visual appearance and skeletal motion provide complementary information for pose

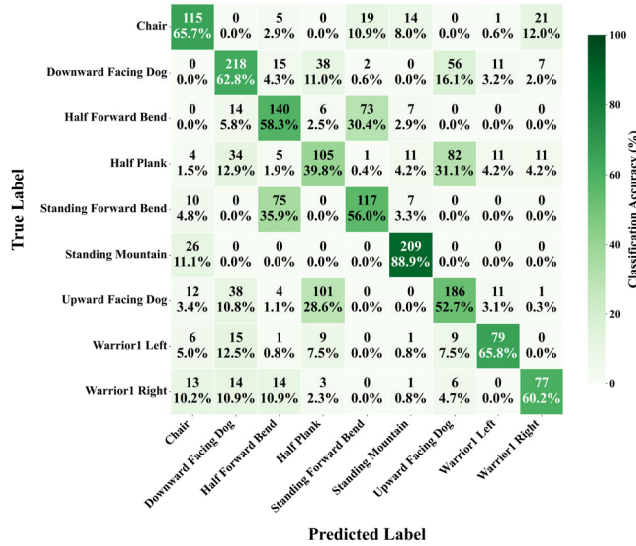


FIGURE 5. Confusion matrix for the fusion (feature-level concatenation) baseline evaluated using 100% labeled training data.

TABLE 9. Pose classification performance of fully supervised end-to-end baselines under different modality configurations. Results are reported as pose-wise mean accuracy and macro-F1 score (%) computed over all leave-one-subject-out (LOSO) test folds, where each pose class contributes equally to the macro-F1 regardless of clip frequency.

Modality	Accuracy (%)	Macro-F1 (%)
RGB	55.19	53.20
Motion	60.84	61.94
Fusion	60.16	61.12
Gated Fusion	62.82	63.71

recognition. In contrast, unimodal baselines exhibit reduced robustness, particularly for poses with higher visual or kinematic similarity.

Figure 5 shows the confusion matrix for the fused supervised baseline under full supervision. Most misclassifications occur between kinematically or visually similar poses, highlighting the limitations of purely supervised end-to-end training in learning well-structured multimodal representations without explicit representation-level constraints.

These supervised results serve as a performance reference. In the following sections, we evaluate how supervised and self-supervised contrastive pretraining improve representation quality, label efficiency, and cross-modal alignment beyond these baselines.

C. SUPERVISED CONTRASTIVE LEARNING (SCL)

SCL plays a central role in the proposed *CoRF-Yoga* framework as the supervised contrastive pretraining strategy, explicitly incorporating pose-level semantic supervision during representation learning. Unlike conventional supervised learning, which optimizes class separation only at the classifier level, SCL directly enforces intra-class compactness and inter-class separability in the embedding space. This property is particularly important for cross-modal yoga pose

TABLE 10. Downstream pose recognition performance reported as pose-wise accuracy (%) under full supervision (100% labeled data). Δ indicates the absolute difference between purely supervised end-to-end training and SCL pretraining.

Modality	Supervised	SCL	Δ
RGB	55.19	99.66	+44.47
Motion	60.84	92.37	+31.53
Fusion (RGB+Motion)	60.16	99.71	+39.55
Gated Fusion (RGB+Motion)	62.82	99.66	+36.84

TABLE 11. RGB-Motion cross-modal retrieval accuracies evaluated in the projection space *after* contrastive pretraining. The random baseline is approximately 11%.

Encoder	Direction	R@1	R@5	R@10	mAP
RGB: ResNet18	RGB→Motion	99.61	99.86	99.90	99.15
Motion: MotionTCN	Motion→RGB	97.44	99.03	99.37	98.42
RGB: ViT-B/16	RGB→Motion	99.71	99.90	99.90	99.04
Motion: MotionTCN	Motion→RGB	97.20	99.37	99.52	98.24
RGB: Qwen2.5-VL-3B	RGB→Motion	97.30	99.76	99.90	95.57
Motion: MotionTCN	Motion→RGB	91.98	94.93	95.65	93.39

recognition, where visual appearance and skeletal motion provide complementary yet heterogeneous cues and exhibit distinct noise characteristics.

By aligning samples that share the same pose label while pushing apart samples from different poses, SCL produces semantically structured representations that are more robust to viewpoint changes, execution variability, and modality-specific noise. In the following, we analyze the impact of SCL on downstream pose classification, cross-modal alignment, and label efficiency.

1) DOWNSTREAM POSE CLASSIFICATION (RGB / MOTION / FUSION)

Table 10 compares fully supervised end-to-end training from scratch with SCL under full supervision (100% labeled data). SCL improves downstream accuracy across all three modality configurations, with the largest gain observed for Motion and consistently strong improvements for RGB+Motion fusion, indicating that supervised contrastive pretraining is particularly beneficial for temporal and multimodal representations.

We additionally evaluated stronger RGB encoders, including ViT-B/16 and Qwen2.5-VL-3B. As shown in Table 11, consistent trends were observed across architectures, where multimodal fusion outperforms RGB-only models. These results indicate that the proposed framework generalizes beyond the ResNet-based backbone.

2) STATISTICAL VALIDATION

To assess the reliability of the results, we report mean $\pm$ standard deviation across LOSO folds for all label fractions (10%–100%). Statistical significance is evaluated using paired t-tests between the supervised baseline and the proposed SCL approach. The corresponding 95% confidence

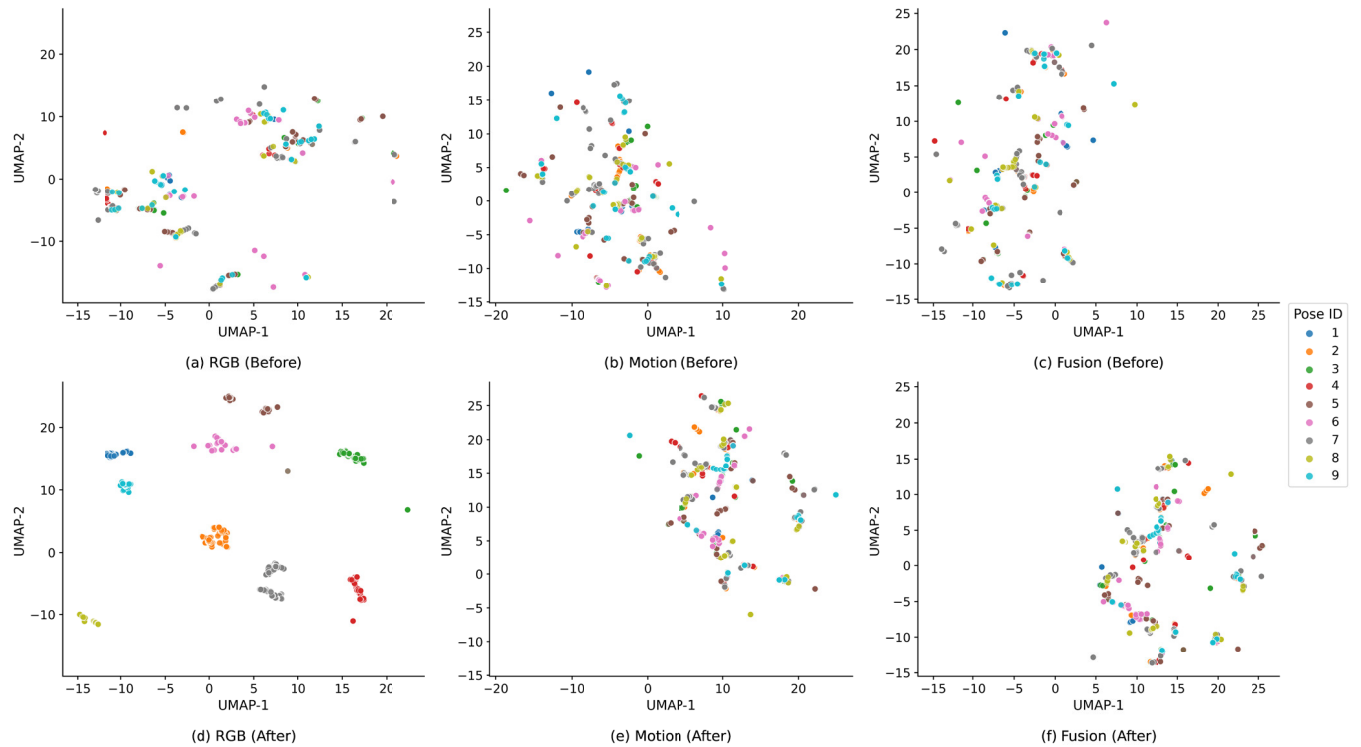


FIGURE 6. UMAP visualization of encoder embeddings before and after supervised contrastive learning (SCL). Top row: representations before contrastive pretraining. Bottom row: representations after contrastive pretraining. Columns correspond to (a,d) RGB, (b,e) Motion, and (c,f) feature-level Fusion. Colors indicate pose IDs: (1) Chair, (2) Downward Facing Dog, (3) Half Forward Bend, (4) Half Plank, (5) Standing Forward Bend, (6) Standing Mountain, (7) Upward Facing Dog, (8) Warrior I-Left, and (9) Warrior I-Right. After contrastive pretraining, intra-class clusters become more compact and inter-class separation becomes clearer across modalities.

TABLE 12. Statistical validation across modalities and label fractions. Results are reported as per-subject mean $\pm$ standard deviation across LOSO folds, with 95% confidence interval half-widths shown in brackets. Statistical significance is evaluated using paired t-tests between the supervised baseline and the proposed SCL approach.

Modality	Method	10%	25%	50%	75%	100%
RGB	Baseline	50.32 $\pm$ 15.73 [10.00]	54.88 $\pm$ 16.64 [10.58]	56.66 $\pm$ 18.27 [11.61]	55.62 $\pm$ 16.96 [10.77]	55.69 $\pm$ 16.54 [10.51]
	SCL	99.73 $\pm$ 0.46 [0.29]	99.69 $\pm$ 0.46 [0.29]	99.77 $\pm$ 0.38 [0.24]	99.77 $\pm$ 0.38 [0.24]	99.72 $\pm$ 0.38 [0.24]
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001
Motion	Baseline	57.13 $\pm$ 22.57 [14.34]	58.10 $\pm$ 21.90 [13.92]	57.97 $\pm$ 22.10 [14.04]	57.66 $\pm$ 24.00 [15.25]	58.73 $\pm$ 22.26 [14.14]
	SCL	80.68 $\pm$ 20.77 [13.20]	84.57 $\pm$ 16.97 [10.78]	88.04 $\pm$ 14.39 [9.14]	89.34 $\pm$ 13.33 [8.47]	89.85 $\pm$ 13.06 [8.30]
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001
Fusion	Baseline	56.07 $\pm$ 15.79 [10.03]	61.28 $\pm$ 19.46 [12.36]	60.33 $\pm$ 18.17 [11.54]	60.12 $\pm$ 17.64 [11.21]	59.73 $\pm$ 17.47 [11.10]
	SCL	99.77 $\pm$ 0.46 [0.29]	99.69 $\pm$ 0.46 [0.29]	99.73 $\pm$ 0.46 [0.29]	99.73 $\pm$ 0.46 [0.29]	99.77 $\pm$ 0.38 [0.24]
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001
Gated Fusion	Baseline	58.88 $\pm$ 20.11 [12.78]	63.42 $\pm$ 21.01 [13.35]	61.43 $\pm$ 18.50 [11.75]	61.17 $\pm$ 18.69 [11.87]	59.30 $\pm$ 18.87 [11.99]
	SCL	99.77 $\pm$ 0.46 [0.29]	99.75 $\pm$ 0.44 [0.28]	99.73 $\pm$ 0.46 [0.29]	99.73 $\pm$ 0.46 [0.29]	99.73 $\pm$ 0.46 [0.29]
	p-value	<0.001	<0.001	<0.001	<0.001	<0.001

intervals are also reported (shown as half-widths) and exhibit consistent trends, further supporting the reliability of the results.

As summarized in Table 12, the differences between SCL and the supervised baseline are statistically significant across all modalities and label fractions ($p < 0.001$). Accordingly, SCL consistently outperforms the baseline in all settings. The baseline models exhibit high variance across subjects

under the LOSO protocol, with standard deviations ranging from 15 to 24, indicating limited cross-subject generalization. In contrast, SCL achieves substantially improved performance with reduced variance, particularly for RGB and fusion-based modalities (often below 0.5), demonstrating both improved accuracy and stability. Notably, SCL maintains high performance even at low label fractions (10%), highlighting its effectiveness in label-efficient learning.

3) CROSS-MODAL ALIGNMENT ANALYSIS

To examine how SCL affects cross-modal representation alignment, we visualize the learned latent spaces and evaluate pose-level cross-modal retrieval. Retrieval is performed immediately after contrastive pretraining using frozen encoders and projection heads, without involving downstream classifier training. Retrieval evaluation is conducted using all labeled pose samples in the DSZ-Yoga dataset.

Figure 6 shows that SCL yields well-separated pose clusters across modalities. Table 11 confirms strong RGB↔Motion alignment after pretraining, supporting the downstream fusion gains.

4) LABEL-EFFICIENCY ANALYSIS

We next analyze the label efficiency of SCL by varying the fraction of labeled training data under the LOSO protocol. Table 12 reports per-subject performance across label fractions (10%–100%), including mean $\pm$ standard deviation and statistical significance for purely supervised training and SCL.

As shown in Table 12, SCL substantially improves label efficiency across all modality configurations. Performance improves rapidly as label availability increases, with strong and statistically consistent gains across subjects, particularly in the fused RGB+Motion setting. These results indicate that most pose-discriminative structure is captured during supervised contrastive pretraining, significantly reducing the amount of labeled data required for high-accuracy multimodal pose recognition.

5) FAILURE CASE ANALYSIS

Figure 7 presents the corresponding confusion matrices across label fractions from 10% to 100%. The results reveal distinct patterns of misclassification across modalities. In the RGB-only setting, confusion mainly occurs between *Half Plank* and *Upward Facing Dog*, and between *Upward Facing Dog* and *Downward Facing Dog*, due to strong visual similarity and the absence of temporal information.

In the motion-only setting, confusion is more prominent between symmetric or structurally similar poses, such as *Warrior I Left* and *Warrior I Right*, as well as between *Downward Facing Dog* and *Upward Facing Dog*, due to motion ambiguity and limited spatial orientation cues.

The fusion model reduces these errors across label fractions by leveraging complementary spatial and temporal information, resulting in substantially improved classification performance. Remaining errors are minimal and do not exhibit consistent patterns.

Failures are primarily caused by visual similarity, motion ambiguity, symmetry, and viewpoint limitations, highlighting the effectiveness of multimodal fusion in improving robustness.

Overall, these results demonstrate that SCL substantially enhances downstream accuracy, cross-modal alignment, and label efficiency, establishing it as a key component of

TABLE 13. Comparison of supervised training from scratch, SCL, and Self-SCL for downstream pose recognition using fused RGB+Motion features, reported as pose-wise accuracy (%).

Method	25% Labels	100% Labels
Supervised (Scratch)	63.25	60.16
SCL	99.61	99.71
Self-SCL	66.10	63.74

TABLE 14. RGB–Motion cross-modal retrieval performance evaluated in the projection space z . Metrics include Recall@K (R@1, R@5, R@10) and mean average precision (mAP). Retrieval is conducted over the full dataset, where each sample is used as a query and the remaining samples form the gallery.

Method	Direction	R@1	R@5	R@10	mAP
SCL (z)	RGB→Motion	99.61	99.86	99.90	99.15
	Motion→RGB	97.44	99.03	99.37	98.42
Self-SCL (z)	RGB→Motion	83.78	96.09	98.07	27.10
	Motion→RGB	85.42	95.85	98.50	27.22

the CoRF-Yoga framework. We next evaluate a label-free alternative using Self-SCL to highlight the trade-off between annotation cost and downstream performance.

D. SELF-SUPERVISED CONTRASTIVE LEARNING (SELF-SCL)

Self-SCL is optimized using the InfoNCE loss to learn representations without pose labels. Unlike SCL, Self-SCL relies on instance-level discrimination and cross-modal correspondence, making it a practical alternative when annotation is limited or unavailable. In the following, we analyze its downstream performance, cross-modal alignment, and label-efficiency trade-offs relative to SCL.

1) DOWNSTREAM POSE CLASSIFICATION PERFORMANCE

As shown in Table 13, Self-SCL improves over purely supervised training under both label regimes, while remaining inferior to SCL. These results highlight the trade-off between annotation cost and downstream accuracy, where SCL achieves the highest performance when labels are available, while Self-SCL recovers a substantial portion of the performance gain without requiring pose annotations during pretraining, making it particularly attractive when labeled data is scarce.

2) CROSS-MODAL REPRESENTATION ALIGNMENT

Figure 8 shows that Self-SCL learns meaningful cross-modal structure but produces less compact pose clusters than SCL. Table 14 supports this observation, indicating weaker alignment without label-informed positive pairs.

3) FAILURE CASE ANALYSIS

Figure 7 presents the corresponding confusion matrices for 25% and 100% label fractions. The results reveal consistent

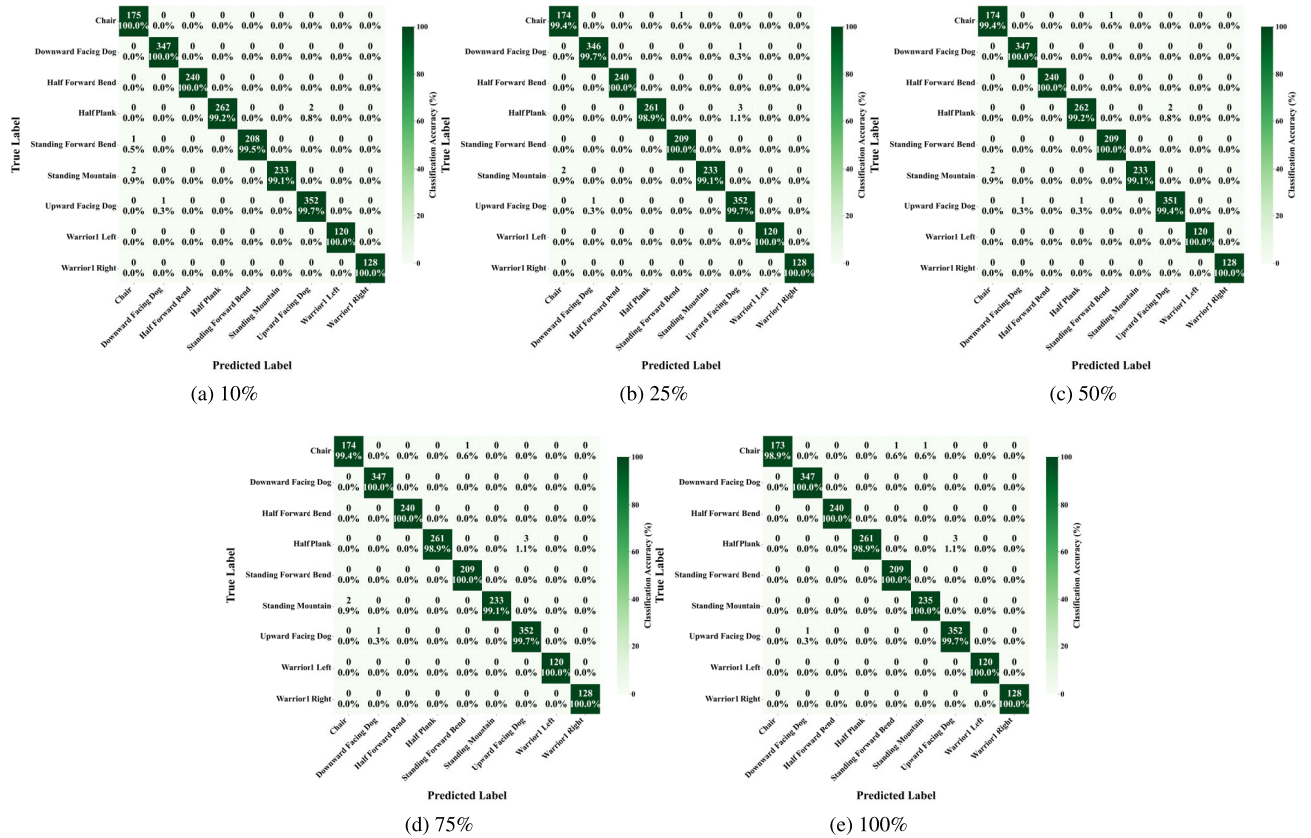


FIGURE 7. Confusion matrices of the downstream fusion classifier using feature concatenation under SCL, evaluated at labeled data fractions ranging from 10% to 100%.

TABLE 15. Per-class precision (P), recall (R), F1-score (F1), and support (S) for fused RGB+Motion models under purely supervised training, SCL and Self-SCL.

Class	Supervised (Scratch)				SCL				Self-SCL			
	P	R	F1	S	P	R	F1	S	P	R	F1	S
Chair	61.83	65.71	63.71	175	100.00	98.86	99.43	175	73.26	78.29	75.69	175
Downward Facing Dog	65.47	62.82	64.12	347	99.71	100.00	99.86	347	62.26	55.62	58.75	347
Half Forward Bend	54.05	58.33	56.11	240	100.00	100.00	100.00	240	58.01	55.83	56.90	240
Half Plank	40.08	39.77	39.92	264	100.00	98.86	99.43	264	55.97	51.52	53.65	264
Standing Forward Bend	55.19	55.98	55.58	209	99.52	100.00	99.76	209	56.73	56.46	56.59	209
Standing Mountain	83.60	88.94	86.19	235	99.58	100.00	99.79	235	77.94	93.19	84.88	235
Upward Facing Dog	54.87	52.69	53.76	353	99.15	99.72	99.44	353	61.93	61.76	61.84	353
Warrior I (Left)	69.91	65.83	67.81	120	100.00	100.00	100.00	120	52.35	65.00	57.99	120
Warrior I (Right)	65.81	60.16	62.86	128	100.00	100.00	100.00	128	79.09	67.97	73.11	128

misclassification patterns across label fractions, with the most frequent confusion occurring between *Half Forward Bend* and *Standing Forward Bend* due to their strong visual similarity and subtle posture differences.

Additional confusion occurs among *Downward Facing Dog*, *Upward Facing Dog*, and *Half Plank*, driven by motion ambiguity and similar global structures, where distinctions rely on orientation and temporal cues.

Symmetric poses such as *Warrior I Left/Right* also exhibit confusion due to mirrored configurations, while minor errors between *Chair* and *Standing Mountain* reflect partial

similarity in upright postures. Some errors are further influenced by self-occlusion and viewpoint limitations, which obscure key joints.

Overall, failures are mainly caused by visual similarity, motion ambiguity, and occlusion, highlighting the limitations of Self-SCL without label supervision.

4) LABEL-EFFICIENCY AND ANNOTATION TRADE-OFFS

Despite achieving lower absolute accuracy than SCL, Self-SCL offers a key advantage in annotation efficiency by enabling representation learning without pose labels during

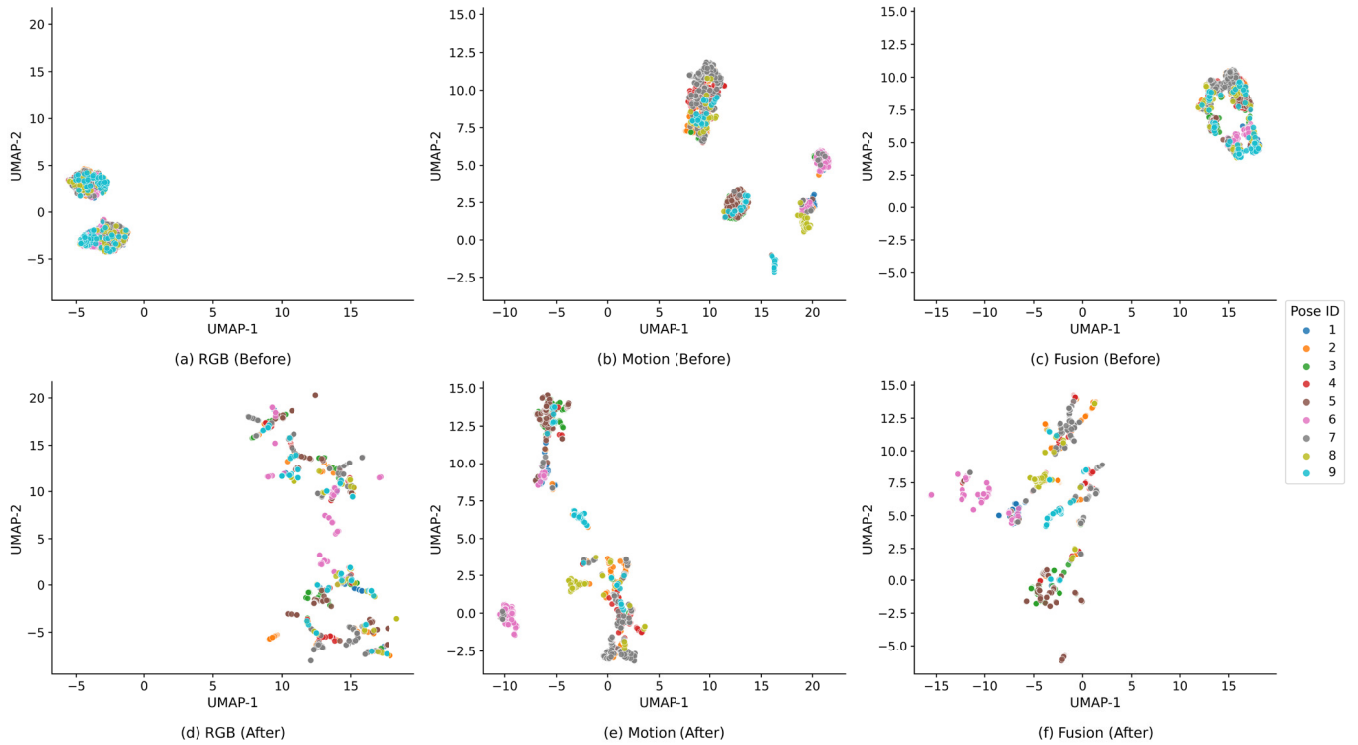


FIGURE 8. UMAP visualization of frozen encoder representations learned via Self-SCL for (a) RGB, (b) Motion, and (c) feature-level Fusion. Colors indicate pose IDs: (1) Chair, (2) Downward Facing Dog, (3) Half Forward Bend, (4) Half Plank, (5) Standing Forward Bend, (6) Standing Mountain, (7) Upward Facing Dog, (8) Warrior I-Left, and (9) Warrior I-Right.

pretraining. In low-label regimes, Self-SCL consistently improves performance over training from scratch, making it particularly attractive for early-stage data collection or large-scale deployments where annotation cost is prohibitive.

From a system design perspective, these results highlight a complementary relationship between Self-SCL and SCL. Self-SCL is well suited for scenarios with limited or unavailable labels, such as incremental dataset expansion or privacy-sensitive environments, while SCL can be introduced once a modest amount of labeled data becomes available to further refine representation quality.

E. CLASS-WISE PERFORMANCE COMPARISON

Table 15 reports per-class results for the fused RGB+Motion models. SCL is the most consistent across poses, while Self-SCL improves over supervised training but remains less stable for similar poses.

F. COMPUTATIONAL EFFICIENCY AND PRACTICAL CONSIDERATIONS

1) TRAINING AND INFERENCE COSTS

We focus on SCL and Self-SCL in this analysis, as both are core components of the proposed CoRF-Yoga framework. Although all methods share the same fused inference architecture, empirical efficiency differs across training strategies when considering both model-only forward latency and end-to-end (E2E) inference time, which includes RGB

decoding, motion loading, preprocessing, and model inference. Detailed results are provided in Table 16. E2E latency is primarily dominated by data processing overhead, with SCL consistently achieving lower latency and higher throughput than Self-SCL across all label fractions. These findings demonstrate that the proposed framework is practical for real-world deployment, as SCL achieves low end-to-end latency (~ 12 ms per sample) and high throughput (~ 83 samples/s), enabling efficient near real-time yoga pose recognition under realistic data processing conditions. All experiments were conducted on the hardware configuration summarized in Table 8.

2) DEPLOYMENT GUIDANCE

Based on the computational efficiency results, CoRF-Yoga supports flexible deployment across a range of hardware settings. All methods share the same fused inference architecture, resulting in comparable parameter counts. However, models pretrained with SCL or Self-SCL consistently achieve lower inference latency and reduced peak memory usage compared to purely supervised baselines.

In scenarios where labeled data is scarce or annotation cost is prohibitive, Self-SCL provides a practical solution by enabling label-free representation learning while maintaining efficient inference-time performance. When a modest amount of labeled data is available, SCL is preferred, as it delivers higher recognition accuracy and more stable cross-modal

TABLE 16. Computational efficiency of fused RGB+Motion models under end-to-end supervised training and frozen-encoder downstream evaluation after SCL and Self-SCL. Training time includes end-to-end optimization for supervised baselines and downstream classifier training for contrastive methods; contrastive pretraining time is excluded. End-to-end (E2E) latency includes RGB decoding/loading, motion loading, preprocessing, and model inference.

Label %	Method	Train Time (hrs)	Forward (ms)	E2E (ms/sample)	Throughput (samples/s)	Params (M)	Peak Memory (GB)
10%	Supervised	0.2985	2.6351	27.3667 ± 22.7717	36.5407	11.9728	0.9479
	SCL	0.1970	4.5521	12.1920 ± 0.7862	82.0207	12.4044	0.3335
	Self-SCL	0.0256	3.7111	19.7164 ± 0.8694	50.7192	12.4044	0.3335
25%	Supervised	0.3048	4.8898	27.3403 ± 22.7212	36.5760	11.9728	0.9473
	SCL	0.2024	4.1595	11.9987 ± 0.8707	83.3426	12.4044	0.3350
	Self-SCL	0.0393	4.1583	19.8335 ± 0.9674	50.4197	12.4044	0.3350
50%	Supervised	0.3149	5.1208	27.4023 ± 22.8480	36.4933	11.9728	0.9496
	SCL	0.2122	3.0494	12.1671 ± 1.0876	82.1887	12.4044	0.3346
	Self-SCL	0.0635	5.0225	19.6727 ± 0.8979	50.8320	12.4044	0.3346
75%	Supervised	0.3258	4.5422	27.4338 ± 22.8145	36.4514	11.9728	0.9505
	SCL	0.2226	4.5647	11.9699 ± 0.8311	83.5429	12.4044	0.3336
	Self-SCL	0.0882	4.2654	19.6415 ± 0.9388	50.9127	12.4044	0.3336
100%	Supervised	0.3364	4.2541	27.5378 ± 22.9368	36.3137	11.9728	0.9485
	SCL	0.2325	2.9632	11.9504 ± 0.8357	83.6791	12.4044	0.3334
	Self-SCL	0.1120	4.4822	19.7698 ± 0.9423	50.5822	12.4044	0.3334

representations while preserving low inference latency and memory usage. Together, these properties make contrastive pretraining well suited for deployment in intelligent environments with real-time and resource constraints.

VI. DISCUSSION

This section interprets the experimental findings with a focus on the central comparisons in this work, namely supervised versus self-supervised cross-modal contrastive learning and their implications for label efficiency in multimodal yoga pose recognition.

A. CROSS-MODAL REPRESENTATION LEARNING

The results demonstrate that contrastive pretraining is critical for learning a shared representation space between RGB visual appearance and skeletal motion. SCL is particularly effective, as pose labels explicitly enforce semantic consistency across modalities, leading to compact and well-aligned embeddings. This structured alignment enables effective feature-level fusion using simple concatenation, without requiring additional fusion parameters.

In contrast, Self-SCL captures meaningful cross-modal structure through temporal synchronization alone but produces less compact pose clusters, reflecting the absence of label-informed positive pairs. This difference directly explains the observed performance gap between SCL and Self-SCL in downstream classification and cross-modal retrieval.

B. ANNOTATION EFFICIENCY AND TRAINING STRATEGY

A key finding is that contrastive pretraining enables flexible training strategies under varying annotation availability. When pose labels are scarce or unavailable, Self-SCL provides a strong label-free initialization that consistently

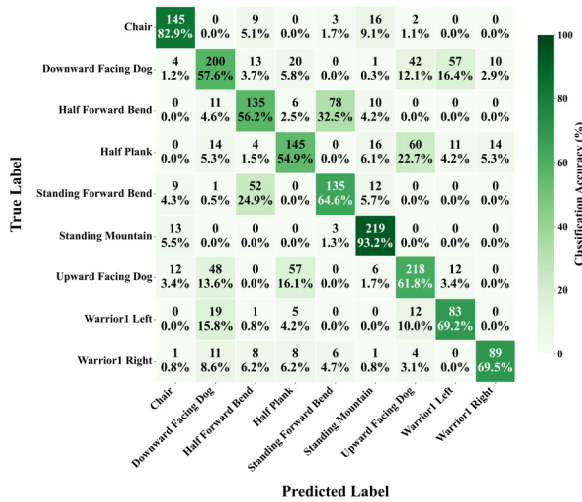
outperforms training from scratch. As soon as a modest amount of labeled data becomes available, SCL rapidly refines the representation space, achieving near-saturated performance with substantially reduced annotation effort.

These results clarify the trade-off between annotation cost and achievable accuracy. Self-SCL is well suited for early-stage data collection or large-scale deployments where labeling is expensive, while SCL offers maximal performance once limited supervision can be introduced.

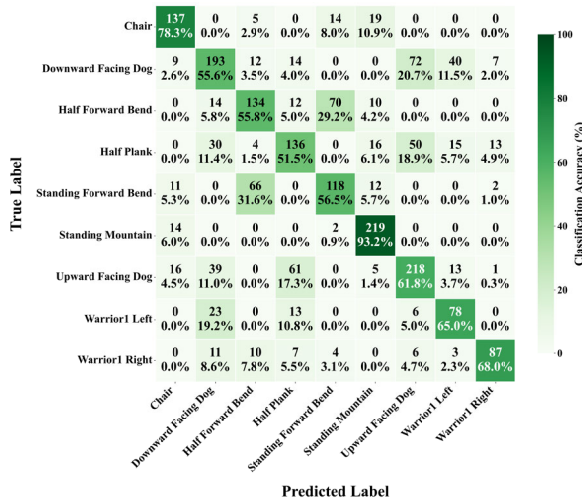
C. PRACTICAL DEPLOYMENT CONSIDERATIONS

Importantly, the performance gains from contrastive learning are achieved without increasing inference-time complexity. All methods share the same encoder and classifier architectures at deployment, while contrastive approaches rely on an offline pretraining stage. Empirical results show that frozen-encoder models achieve lower inference latency and reduced memory usage compared to end-to-end supervised training, making contrastive pretraining compatible with resource-constrained and privacy-aware intelligent environments.

Existing vision-motion cross-modal approaches often employ more complex architectures such as GCNs or Transformers, which typically require larger datasets and higher computational costs. In contrast, CoRF-Yoga adopts a lightweight Temporal Convolutional Network (TCN) for motion encoding. This design is motivated by its computational efficiency, suitability for real-time intelligent environments, and improved generalization under the relatively small DSZ-Yoga dataset. Moreover, using a simple encoder allows the study to focus on the effectiveness of the proposed cross-modal contrastive learning framework, rather than confounding performance gains with increased backbone complexity.



(a) 25% labeled data



(b) 100% labeled data

FIGURE 9. Confusion matrices for the fusion (feature-level concatenation) Self-SCL model evaluated at 25% and 100% labeled training data.

In addition, the strong performance of motion-only and motion-enhanced models highlights the practical value of skeletal motion as a sensing modality. Motion representations remain informative even when visual cues are degraded or unavailable, indicating that accurate pose recognition can be achieved using motion data alone, as supported by the experimental results.

D. LIMITATIONS

This study has several limitations. The dataset includes a limited number of participants. Although a leave-one-subject-out (LOSO) protocol is adopted to evaluate cross-subject generalization, the limited number of participants may constrain the robustness and generalizability of the results. This limitation can be addressed by expanding the dataset to include a larger and more diverse participant pool.

The dataset also exhibits demographic limitations, including gender imbalance and limited diversity in yoga

experience, which may bias the learned representations toward the dominant participant group and limit generalization to populations with different body characteristics, flexibility levels, and execution styles. A more demographically balanced dataset would help mitigate this limitation. In addition, experiments rely on a single RGB viewpoint and simplified short-term temporal modeling. Incorporating multi-view RGB data and more expressive temporal modeling would improve the capture of motion dynamics.

Another limitation arises from asymmetric encoder initialization in the baseline comparisons. The RGB encoder is initialized with ImageNet-pretrained weights, whereas the motion encoder is trained from scratch due to the lack of large-scale pretrained models for skeletal motion data. This difference may introduce modality bias, potentially affecting the fairness of cross-modal comparisons. Although both encoders are jointly optimized, this asymmetry may still influence the relative performance across modalities. Developing or leveraging large-scale pretraining strategies for motion encoders would help reduce modality bias and enable more balanced cross-modal comparisons.

A further limitation is the absence of external validation on existing public benchmarks. The proposed dataset contains synchronized RGB and IMU-based motion data, while public datasets differ in modality and label space. Direct comparison would require reducing the framework to a single-modality setting (e.g., RGB-only), which is not directly comparable to the proposed cross-modal design. Therefore, cross-dataset generalization remains an open challenge and an important direction for future work. Developing compatible cross-modal benchmarks with synchronized RGB and motion data would enable a more direct assessment of this challenge.

An additional limitation is that the current self-supervised evaluation is limited in scope and does not include a comprehensive comparison with standard SSL methods (e.g., SimCLR, BYOL, VICReg) under a unified frozen-encoder evaluation protocol. This remains an important direction for future work.

Furthermore, the dataset focuses on a single, well-defined yoga sequence (Sun Salutation B), which limits pose diversity and movement variability. Addressing this limitation will require larger and more diverse datasets, including a broader range of yoga poses and sequences, and richer temporal representations that explicitly model pose transitions and continuous movement dynamics.

VII. CONCLUSION AND FUTURE WORK

This paper introduced DSZ-Yoga, a synchronized RGB-motion dataset for fine-grained yoga pose recognition, and proposed *CoRF-Yoga*, a two-stage cross-modal contrastive learning framework for label-efficient multimodal representation learning. By supporting both SCL and Self-SCL within an identical architectural pipeline, the framework enables a controlled investigation of how supervision level influences cross-modal representation learning.

TABLE 17. SCL Configuration.

ID	Pretraining	Head Dim	τ	Clip Length	Sampling Rate	Purpose
A1	No-contrastive	N/A	N/A	16	2	No-contrastive baseline
A2	SCL	128	0.07	16	2	SCL reference setting
A3	SCL	64	0.07	16	2	Reduced projection head dimension
A4	SCL	256	0.07	16	2	Increased projection head dimension
A5	SCL	128	0.03	16	2	Lower temperature τ
A6	SCL	128	0.1	16	2	Higher temperature τ
A7	SCL	128	0.07	8	2	Reduced temporal window (clip length)
A8	SCL	128	0.07	32	2	Extended temporal window (clip length)
A9	SCL	128	0.07	16	1	Lower temporal sampling rate
A10	SCL	128	0.07	16	4	Higher temporal sampling rate

TABLE 18. Ablation results of SCL under different configurations across varying label fractions. Results are averaged over three representative LOSO subjects.

ID	10%	25%	50%	75%	100%	AvgAcc
A1	49.36	50.67	44.51	48.68	46.79	48.00
A2	97.14	97.25	95.60	95.85	95.92	96.35
A3	99.75	99.75	99.75	99.75	99.75	99.75
A4	99.61	99.61	99.61	99.61	99.61	99.61
A5	99.75	99.75	99.75	99.75	99.75	99.75
A6	99.61	99.61	99.61	99.61	99.61	99.61
A7	96.10	96.00	96.04	97.18	97.72	96.61
A8	99.75	99.75	99.75	99.75	99.75	99.75
A9	99.75	99.49	99.61	99.75	99.75	99.67
A10	97.89	95.79	98.68	97.81	97.78	97.59

TABLE 19. Macro-F1 scores corresponding to the ablation configurations in Table 18.

ID	10%	25%	50%	75%	100%	Avg. Macro-F1
A1	46.92	42.45	41.25	45.04	42.99	43.73
A2	93.52	95.56	94.97	95.11	92.74	94.38
A3	96.17	96.17	96.17	96.17	96.17	96.17
A4	96.06	96.06	96.06	96.06	96.06	96.06
A5	98.63	96.17	96.17	96.17	96.17	96.66
A6	96.06	96.06	96.06	96.06	96.06	96.06
A7	93.13	92.66	96.09	96.00	96.50	94.88
A8	96.17	96.17	96.17	96.17	96.17	96.17
A9	96.17	96.03	96.06	96.17	96.17	96.12
A10	94.24	95.38	94.91	96.63	96.55	95.54

Experimental results indicate that contrastive pretraining facilitates effective cross-modal representation alignment and supports competitive performance under limited annotation settings.

In particular, the findings highlight the role of supervised contrastive learning in enhancing representation quality, while also revealing the limitations of fully self-supervised approaches in fine-grained pose discrimination. These observations provide insight into the trade-offs between supervision and label efficiency in multimodal learning.

Rather than suggesting absolute performance superiority, the results suggest that cross-modal contrastive learning offers a promising direction for reducing annotation requirements while maintaining strong performance within the evaluated setting. The lightweight design and consistent inference pipeline further support its applicability in practical and resource-constrained environments.

Several directions for future work can further extend this framework. Future work will explore pretraining strategies for motion encoders to enable more balanced and fair baseline comparisons in multimodal evaluation. Future studies will evaluate subject-wise or session-wise generalization protocols to better quantify performance on unseen users, and will exploit additional RGB viewpoints to improve robustness against viewpoint changes. Incorporating richer temporal modeling for RGB may further reduce confusion among visually similar poses. Expanding the dataset to include more diverse participants, environments, and execution styles will strengthen ecological validity. In addition, extending the framework to additional sensing modalities and studying robustness under systematic modality dropout are promising directions. Finally, benchmarking inference on resource-constrained devices and exploring model compression techniques will support real-world edge deployment.

Overall, DSZ-Yoga and CoRF-Yoga provide a structured framework for studying cross-modal representation learning, offering empirical insights into how contrastive learning strategies influence performance and label efficiency for realistic multimodal yoga pose recognition.

CODE AVAILABILITY

To ensure reproducibility, the implementation code used in this study is publicly available on GitHub at <https://github.com/dzolboo/CoRF-Yoga>

APPENDIX

ADDITIONAL ABLATION STUDIES

We conduct a comprehensive ablation study to systematically evaluate the impact of key design components in both the SCL and Self-SCL frameworks.

TABLE 20. Self-SCL configuration.

ID	Pretraining	τ	λ	Schedule	Clip Length	Sampling Rate	Purpose
B1	Self-SCL	0.10	1.0	No warm-up	16	2	Baseline (final configuration)
B2	Self-SCL	0.10	0.0	No warm-up	16	2	No intra-modal loss
B3	Self-SCL	0.10	0.1	No warm-up	16	2	Lower λ
B4	Self-SCL	0.10	0.5	No warm-up	16	2	Medium λ
B5	Self-SCL	0.10	1.0	Stepwise warm-up	16	2	Proposed λ schedule

TABLE 21. Ablation results of Self-SCL across label fractions under full 12-subject LOSO evaluation.

ID	10%	25%	50%	75%	100%	Avg. Acc.
B1	68.86	66.10	65.72	64.56	63.74	65.79
B2	65.09	64.22	65.09	64.99	61.52	64.18
B3	66.39	65.86	62.43	65.14	63.64	64.69
B4	68.08	66.15	62.24	62.53	61.81	64.16
B5	69.24	65.09	62.92	64.08	62.34	64.73

TABLE 22. Macro-F1 scores corresponding to the ablation configurations in Table 21.

ID	10%	25%	50%	75%	100%	Avg. Macro-F1
B1	68.97	66.76	66.49	65.54	64.38	66.43
B2	65.92	65.60	66.93	65.78	62.64	65.37
B3	66.10	66.86	63.65	65.65	64.38	65.33
B4	67.93	67.36	63.60	63.52	62.18	64.92
B5	69.65	66.83	63.86	64.90	63.47	65.74

- 1) **Pretraining strategy:** Comparison between no contrastive pretraining (A1), Self-SCL (default configuration B1), and the proposed SCL (A2).
- 2) **Projection head dimension:** Evaluation of different projection dimensions (A3–A4).
- 3) **Temperature parameter (τ):** Analysis of different τ values (A5–A6).
- 4) **Temporal design (clip length and sampling rate):** Evaluation of the effects of clip length (A7–A8) and sampling rate (A9–A10) on downstream performance.
- 5) **Intra-modal loss weight (λ):** Since λ is specific to the Self-SCL objective, we conduct a separate ablation under the Self-SCL setting (B1–B5) to analyze its effect, including removal of the intra-modal loss ($\lambda = 0$), reduced and moderate values, and a stepwise warm-up schedule.

To ensure a feasible evaluation given the large number of ablation configurations while maintaining representativeness, ablations (i)–(iv) are conducted on a subset of three LOSO subjects selected to reflect varying levels of difficulty (easy, moderate, and challenging), as identified from the full LOSO evaluation. These subset-based results are used to analyze configuration trends. In contrast, ablation (v) and all primary results in the manuscript are conducted on the full 12-subject LOSO evaluation.

Table 17 summarizes the ablation configurations used to systematically evaluate the impact of pretraining strategy, projection head dimension, temperature τ , and temporal design (clip length and sampling rate) under the SCL framework.

The ablation results confirm that contrastive pretraining is essential, as the no-contrastive baseline (A1) performs substantially worse than the SCL reference setting (A2) across all label fractions (Table 18). Performance is generally robust across configurations, although smaller projection dimension and lower temperature yield slightly better results in this subset (Table 18). Temporal design also affects performance: reducing clip length degrades results, while longer clips improve performance (Table 18). For sampling rate, lower sampling rates (denser temporal sampling) yield better performance, whereas higher sampling rates lead to slight degradation (Table 18). Overall, these findings support the effectiveness of the SCL framework and highlight the importance of temporal design choices for downstream fusion performance (Tables 18 and 19).

In addition to the SCL ablation, we further investigate the Self-SCL objective to provide a comprehensive evaluation of its design components.

For the LOSO setting, the Self-SCL training recipe was refined to improve optimization stability, including stronger multimodal augmentations, a revised projection head, and stabilized optimization settings. All ablation experiments are conducted using this final configuration, while varying only the intra-modal loss weight λ and its scheduling strategy.

This ablation study examines the impact of the intra-modal loss weight λ and its scheduling within the Self-SCL framework, directly addressing their role in the overall training objective (Tables 20–22).

The study compares (i) removal of the intra-modal loss ($\lambda = 0$), (ii) reduced and moderate λ values, and (iii) a stepwise warm-up schedule, while keeping all other factors fixed.

The results indicate that the baseline configuration with full intra-modal weighting ($\lambda = 1.0$, B1) achieves the best average performance across label fractions under the full 12-subject LOSO evaluation (Tables 21 and 22). Reducing or removing the intra-modal loss (B2–B3) does not consistently improve overall performance, while moderate weighting (B4) yields comparable but not superior results. In addition, the stepwise warm-up strategy (B5) does not provide further

gains and slightly degrades performance compared to a fixed λ setting.

These observations suggest that while intra-modal regularization is not strictly necessary, it contributes positively when appropriately balanced with the cross-modal objective. In contrast, the warm-up scheduling strategy is not essential in this setting. Overall, cross-modal contrastive alignment remains the primary driver of representation learning, with intra-modal loss acting as a secondary, data-dependent regularizer.

For completeness and consistency with the overall framework, we retain the full Self-SCL formulation in the main experiments, and use this ablation study to clarify the contribution of each component.

REFERENCES

- [1] J.-S. Tan, B. K. Beheshti, T. Binnie, P. Davey, J. P. Caneiro, P. Kent, A. Smith, P. O'Sullivan, and A. Campbell, "Human activity recognition for people with knee osteoarthritis—A proof-of-concept," *Sensors*, vol. 21, no. 10, p. 3381, May 2021, doi: [10.3390/s21103381](https://doi.org/10.3390/s21103381).
- [2] K. Host and M. Ivačić-Kos, "An overview of human action recognition in sports based on computer vision," *Heliyon*, vol. 8, no. 6, Jun. 2022, Art. no. e09633, doi: [10.1016/j.heliyon.2022.e09633](https://doi.org/10.1016/j.heliyon.2022.e09633).
- [3] B. Yu, Y. Liu, and K. Chan, "A survey of sensor modalities for human activity recognition," in *Proc. 12th Int. Joint Conf. Knowl. Discovery, Knowl. Eng. Knowl. Manage.*, 2020, pp. 282–294, doi: [10.5220/0010145202820294](https://doi.org/10.5220/0010145202820294).
- [4] J. Carreira, E. Noland, C. Hillier, and A. Zisserman, "A short note on the kinetics-700 human action dataset," 2019, *arXiv:1907.06987*.
- [5] J. R. Kwapisz, G. M. Weiss, and S. A. Moore, "Activity recognition using cell phone accelerometers," *ACM SIGKDD Explor. Newslett.*, vol. 12, no. 2, pp. 74–82, Mar. 2011, doi: [10.1145/1964897.1964918](https://doi.org/10.1145/1964897.1964918).
- [6] J. Ni, H. Tang, S. T. Haque, Y. Yan, and A. H. H. Ngu, "A survey on multimodal wearable sensor-based human action recognition," 2024, *arXiv:2404.15349*.
- [7] J. Shin, N. Hassan, A. S. M. Miah, and S. Nishimura, "A comprehensive methodological survey of human activity recognition across diverse data modalities," *Sensors*, vol. 25, no. 13, p. 4028, Jun. 2025.
- [8] M. M. Islam, S. Nooruddin, F. Karay, and G. Muhammad, "Human activity recognition using tools of convolutional neural networks: A state of the art review, data sets, challenges, and future prospects," *Comput. Biol. Med.*, vol. 149, Oct. 2022, Art. no. 106060, doi: [10.1016/j.combiomed.2022.106060](https://doi.org/10.1016/j.combiomed.2022.106060).
- [9] K. P. Jois, *Yoga Mala: The Original Teachings Ashtanga Yoga Master Sri K. Pattabhi Jois*. New York, NY, USA: North Point Press, 2010.
- [10] M. Verma, S. Kumawat, Y. Nakashima, and S. Raman, "Yoga-82: A new dataset for fine-grained classification of human poses," 2020, *arXiv:2004.10362*.
- [11] Z. Wu, J. Zhang, K. Chen, and C. Fu, "Yoga posture recognition and quantitative evaluation with wearable sensors based on two-stage classifier and prior Bayesian network," *Sensors*, vol. 19, no. 23, p. 5129, Nov. 2019, doi: [10.3390/s19235129](https://doi.org/10.3390/s19235129).
- [12] D. M. Kishore, S. Bindu, and N. K. Manjunath, "Estimation of yoga postures using machine learning techniques," *Int. J. Yoga*, vol. 15, no. 2, pp. 137–143, May 2022, doi: [10.4103/ijoy.ijoy_97_22](https://doi.org/10.4103/ijoy.ijoy_97_22).
- [13] D. Swain, S. Satapathy, B. Acharya, M. Shukla, V. C. Gerogiannis, A. Kanavos, and D. Giakovis, "Deep learning models for yoga pose monitoring," *Algorithms*, vol. 15, no. 11, p. 403, Oct. 2022, doi: [10.3390/a15110403](https://doi.org/10.3390/a15110403).
- [14] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, "Supervised contrastive learning," in *Proc. 34th Int. Conf. Neural Inf. Process. Syst.* Red Hook, NY, USA: Curran Associates Inc., 2020, pp. 18661–18673.
- [15] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. 37th Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 1597–1607. [Online]. Available: <https://proceedings.mlr.press/v119/chent20j.html>
- [16] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. A. Pires, Z. Guo, and M. G. Azar, "Bootstrap your own latent: A new approach to self-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 21271–21284.
- [17] C. Feichtenhofer, H. Fan, B. Xiong, R. Girshick, and K. He, "A large-scale study on unsupervised spatiotemporal representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3298–3308, doi: [10.1109/CVPR46437.2021.00331](https://doi.org/10.1109/CVPR46437.2021.00331).
- [18] T. Pan, Y. Song, T. Yang, W. Jiang, and W. Liu, "VideoMoCo: Contrastive video representation learning with temporally adversarial examples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11200–11209, doi: [10.1109/CVPR46437.2021.01105](https://doi.org/10.1109/CVPR46437.2021.01105).
- [19] M. Zolfaghari, Y. Zhu, P. Gehler, and T. Brox, "CrossCLR: Cross-modal contrastive learning for multi-modal video representations," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 1430–1439, doi: [10.1109/ICCV48922.2021.00148](https://doi.org/10.1109/ICCV48922.2021.00148).
- [20] J. Zhang, L. Lin, and J. Liu, "Hierarchical consistent contrastive learning for skeleton-based action recognition with growing augmentations," in *Proc. 37th AAAI Conf. Artif. Intell. 35th Conf. Innov. Appl. Artif. Intell. 13th Symp. Educ. Adv. Artif. Intell.*, 2023, pp. 3427–3435, doi: [10.1609/aaai.v37i3.25451](https://doi.org/10.1609/aaai.v37i3.25451).
- [21] J. Kang, J. Shin, J. Shin, D. Lee, and A. Choi, "Robust human activity recognition by integrating image and accelerometer sensor data using deep fusion network," *Sensors*, vol. 22, no. 1, p. 174, Dec. 2021.
- [22] G. Tevet, B. Gordon, A. Hertz, A. H. Bermano, and D. Cohen-Or, "MotionCLIP: Exposing human motion generation to CLIP space," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2022, pp. 358–374, doi: [10.1007/978-3-031-20047-2_21](https://doi.org/10.1007/978-3-031-20047-2_21).
- [23] S. Moon, A. Madotto, Z. Lin, A. Saraf, A. Bearman, and B. Damavandi, "IMU2CLIP: Language-grounded motion sensor translation with multimodal contrastive learning," in *Proc. Findings Assoc. Comput. Linguistics EMNLP*, Dec. 2023, pp. 13246–13253. [Online]. Available: <https://aclanthology.org/2023.findings-emnlp.883/>
- [24] S. G. Fritsch, C. Oguz, V. F. Rey, L. Ray, M. Kiefer-Emmanouilidis, and P. Lukowicz, "MuJo: Multimodal joint feature space learning for human activity recognition," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. (PerCom)*, Mar. 2025, pp. 1–12, doi: [10.1109/PERCOM4205.2025.00020](https://doi.org/10.1109/PERCOM4205.2025.00020).
- [25] R. Pal, D. Adhikari, M. B. B. Heyat, I. Ullah, and Z. You, "Yoga meets intelligent Internet of Things: Recent challenges and future directions," *Bioengineering*, vol. 10, no. 4, p. 459, Apr. 2023, doi: [10.3390/bioengineering10040459](https://doi.org/10.3390/bioengineering10040459).
- [26] A. Ashok, B. Surendiran, and S. Babu, "Yoga pose estimation using MoveNet deep learning models," *Sādhanā*, vol. 50, Jun. 2025, Art. no. 125, doi: [10.1007/s12046-025-02788-w](https://doi.org/10.1007/s12046-025-02788-w).
- [27] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735, doi: [10.1109/CVPR42600.2020.00975](https://doi.org/10.1109/CVPR42600.2020.00975).
- [28] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 12310–12320.
- [29] A. Bardes, J. Ponce, and Y. LeCun, "VICReg: Variance-invariance-covariance regularization for self-supervised learning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022. [Online]. Available: <https://openreview.net/forum?id=xm6YD62D1Ub>
- [30] A. Dutta and A. Zisserman, "The VIA annotation software for images, audio and video," in *Proc. 27th ACM Int. Conf. Multimedia*. New York, NY, USA: ACM, Oct. 2019, pp. 2276–2279, doi: [10.1145/3343031.3350535](https://doi.org/10.1145/3343031.3350535).
- [31] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
- [32] S. Yan, Y. Xiong, and D. Lin, "Spatial temporal graph convolutional networks for skeleton-based action recognition," 2018, *arXiv:1801.07455*.
- [33] L. Shi, Y. Zhang, J. Cheng, and H. Lu, "Two-stream adaptive graph convolutional networks for skeleton-based action recognition," 2018, *arXiv:1805.07694*.
- [34] H. Duan, Y. Zhao, K. Chen, D. Lin, and B. Dai, "Revisiting skeleton-based action recognition," 2021, *arXiv:2104.13586*.
- [35] D. Xie, X. Zhang, X. Gao, H. Zhao, and D. Du, "MAF-Net: A multimodal data fusion approach for human action recognition," *PLoS ONE*, vol. 20, no. 4, pp. 1–27, Apr. 2025, doi: [10.1371/journal.pone.0319656](https://doi.org/10.1371/journal.pone.0319656).



ZOLBOO DAMIRAN (Graduate Student Member, IEEE) received the B.Sc. and M.Sc. degrees in computer science from Mongolian University of Science and Technology, in 2008 and 2010, respectively. She is currently pursuing the Ph.D. degree with the Graduate School of Science and Technology, Nara Institute of Science and Technology (NAIST). She has been a Lecturer with Mongolian University of Science and Technology, since 2010. Her research interests include ubiquitous and IoT-enabled sensing and computing, multimodal human activity recognition, intelligent environments, and artificial intelligence.



HIROHIKO SUWA (Member, IEEE) received the B.E. and M.E. degrees from Gunma University, in 1998, and the Ph.D. degree from The University of Electro-Communications, Japan, in 2000 and 2006, respectively. He was an Assistant Professor with The University of Electro-Communications, from 2006 to 2014, and Nara Institute of Science and Technology (NAIST), from 2014 to 2018, where he has been an Associate Professor, since 2019. His research interests include social informatics, real data mining, and ubiquitous computing systems. He is a member of IEICE and IPSJ.



TOMOKAZU MATSUI (Member, IEEE) received the B.E. degree from the National Institute of Technology, Nara College, in 2019, and the M.E. and Ph.D. degrees from Nara Institute of Science and Technology (NAIST), Nara, Japan, in 2021 and 2023, respectively. Currently, he is an Assistant Professor with the Graduate School of Science and Technology, NAIST. His research interests include ubiquitous computing, smart home, activity recognition, and elderly monitoring. He is a member of IPSJ.



YUKI MATSUDA (Member, IEEE) was born in 1993. He received the B.E. degree in the advanced course of mechanical and electronic system engineering from the National Institute of Technology, Akashi College, Japan, in 2015, and the M.E. and Ph.D. degrees from the Graduate School of Information Science, Nara Institute of Science and Technology, Japan, in 2016 and 2019, respectively. He is currently a Lecturer with the Faculty of Environmental, Life, Natural Science and Technology, Okayama University, Japan. His current research interests include participatory sensing, location-based information systems, wearable computing, and affective computing. He is a member of IPSJ, IEICE, JSAI, and ACM.



KEIICHI YASUMOTO (Senior Member, IEEE) received the B.E., M.E., and Ph.D. degrees in information and computer sciences from The University of Osaka, Osaka, Japan, in 1991, 1993, and 1996, respectively. He is currently a Professor at the Graduate School of Science and Technology, Nara Institute of Science and Technology. His research interests include distributed systems, mobile computing, and ubiquitous computing. He is a member of ACM, IPSJ, and IEICE.

...